

Perioperative infection prediction -- subgroup deployment-readiness audit (ascertainment window (2018–2022))

Model	Random forest (M2, end of surgery), published out-of-fold predictions
Cohort	Inselspital Bern -- ascertainment window (2018–2022): n=6,051; 501 events
Outcome	Present-on-Admission-coded postoperative bacterial infection
Validation	5-fold stratified CV with SMOTE inside training folds (OOF predictions)
Threshold	0.1 (reference operating anchor, not an elicited optimum)
Uncertainty	2,000 patient-level bootstrap resamples; pre-registered event floors applied (AUROC 25, calibration intercept 25, calibration slope 100, net benefit 50)
Mitigation	Not shown here. The mitigation analysis is the cross-fitted recalibration ladder reported in Table 4 and Supplementary Figure 1; a split-based before/after would be a different analysis.
Window	opyear >= 2018

Audit summary

N: 6051	Events: 501	Prevalence: 0.083	Threshold: 0.100
Bootstrap resamples: 2000	Resampling: subgroup-stratified patient-level	Attributes: 9	

Cohort characterisation (Table 1)

6 of 8 variables differ significantly across 'sex' at FDR $q < 0.05$ (Benjamini–Hochberg).

		Grouped by sex			
		Overall	Female	Male	
n		6051	2090	3961	
age, median [Q1,Q3]		66.0 [55.0,74.0]	65.0 [53.0,74.0]	66.0 [56.0,74.0]	0.037 0.0493
age_group, n (%)	0-17	223 (3.7)	109 (5.2)	114 (2.9)	<0.001 0.0016
	18-44	518 (8.6)	220 (10.5)	298 (7.5)	
	45-64	2049 (33.9)	683 (32.7)	1366 (34.5)	
	65-79	2698 (44.6)	839 (40.1)	1859 (46.9)	
	80+	563 (9.3)	239 (11.4)	324 (8.2)	
asa, n (%)	1	666 (11.0)	339 (16.2)	327 (8.3)	<0.001 0.0016
	2	113 (1.9)	47 (2.2)	66 (1.7)	
	3	1967 (32.5)	822 (39.3)	1145 (28.9)	
	4	3305 (54.6)	882 (42.2)	2423 (61.2)	
frailty, n (%)	frail	3684 (60.9)	1010 (48.3)	2674 (67.5)	<0.001 0.0016
	non_frail	547 (9.0)	275 (13.2)	272 (6.9)	
	pre_frail	1820 (30.1)	805 (38.5)	1015 (25.6)	
emergencysurgery, n (%)	0	5162 (85.3)	1806 (86.4)	3356 (84.7)	0.085 0.0971
	1	889 (14.7)	284 (13.6)	605 (15.3)	
duration_of_surg, median [Q1,Q3]		217.0 [150.0,288.0]	196.0 [136.0,270.0]	227.0 [160.0,294.0]	<0.001 0.0016
surgical_group, n (%)	chopcluster_1	395 (6.5)	158 (7.6)	237 (6.0)	<0.001 0.0016
	chopcluster_10	186 (3.1)	58 (2.8)	128 (3.2)	
	chopcluster_11	178 (2.9)	178 (8.5)	0 (0.0)	
	chopcluster_12	826 (13.7)	357 (17.1)	469 (11.8)	
	chopcluster_13	154 (2.5)	69 (3.3)	85 (2.1)	
	chopcluster_3	414 (6.8)	155 (7.4)	259 (6.5)	
	chopcluster_4	2245 (37.1)	502 (24.0)	1743 (44.0)	
	chopcluster_5	328 (5.4)	82 (3.9)	246 (6.2)	
	chopcluster_6	373 (6.2)	178 (8.5)	195 (4.9)	

Grouped by sex						
		Overall	Female	Male	P-Value	q_fdr
	chopcluster_7	143 (2.4)	57 (2.7)	86 (2.2)		
	chopcluster_8	142 (2.3)	47 (2.2)	95 (2.4)		
	none	667 (11.0)	249 (11.9)	418 (10.6)		
outcome, n (%)	0	5550 (91.7)	1901 (91.0)	3649 (92.1)	0.129	0.129
	1	501 (8.3)	189 (9.0)	312 (7.9)		

Per-variable effect sizes and BH-FDR

Variable	Kind	n	p-value	q (FDR)	Effect size	Kind	Magnitude
duration_of_surg	continuous	6051	0.001	0.0016	-0.187	d	negligible
frailty	categorical	6051	0.001	0.0016	+0.190	v	small
age_group	categorical	6051	0.001	0.0016	+0.104	v	small
asa	categorical	6051	0.001	0.0016	+0.189	v	small
surgical_group	categorical	6051	0.001	0.0016	+0.308	v	medium
age	continuous	6051	0.037	0.0493	-0.111	d	negligible
emergencysurgery	binary	6051	0.085	0.0971	-0.048	h	negligible
outcome	binary	6051	0.129	0.1290	+0.042	h	negligible

Discrimination

Attribute	Subgroup	n	Events	Prevalence	AUROC (95% CI)	AUPRC (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)
sex	Overall	6051	501	0.083	0.855 (0.836–0.873)	0.440 (0.394–0.486)	0.900 (0.873–0.926)	0.590 (0.576–0.602)
sex	female	2090	189	0.090	0.857 (0.826–0.886)	0.456 (0.385–0.530)	0.905 (0.862–0.944)	0.584 (0.563–0.606)
sex	male	3961	312	0.079	0.853 (0.830–0.876)	0.430 (0.372–0.487)	0.897 (0.862–0.930)	0.592 (0.576–0.608)
age_group	Overall	6051	501	0.083	0.855 (0.836–0.873)	0.440 (0.394–0.486)	0.900 (0.873–0.926)	0.590 (0.576–0.602)
age_group	0-17	223	21	0.094	0.803	0.350	0.857	0.441
age_group	18-44	518	34	0.066	0.823 (0.756–0.884)	0.396 (0.243–0.540)	0.824 (0.692–0.938)	0.620 (0.574–0.662)
age_group	45-64	2049	146	0.071	0.867 (0.831–0.898)	0.427 (0.347–0.511)	0.918 (0.872–0.959)	0.610 (0.588–0.632)
age_group	65-79	2698	247	0.092	0.872 (0.845–0.896)	0.515 (0.450–0.580)	0.915 (0.878–0.948)	0.598 (0.579–0.617)
age_group	80+	563	53	0.094	0.768 (0.699–0.829)	0.288 (0.191–0.411)	0.849 (0.741–0.938)	0.502 (0.457–0.546)
frailty	Overall	6051	501	0.083	0.855 (0.836–0.873)	0.440 (0.394–0.486)	0.900 (0.873–0.926)	0.590 (0.576–0.602)
frailty	frail	3684	399	0.108	0.860 (0.839–0.879)	0.495 (0.442–0.548)	0.927 (0.901–0.952)	0.533 (0.516–0.551)
frailty	non_frail	547	6	0.011	0.840	0.099	0.833	0.782
frailty	pre_frail	1820	96	0.053	0.791 (0.737–0.837)	0.241 (0.168–0.332)	0.792 (0.701–0.868)	0.636 (0.614–0.659)
clinic	Overall	6051	501	0.083	0.855 (0.836–0.873)	0.440 (0.394–0.486)	0.900 (0.873–0.926)	0.590 (0.576–0.602)
clinic	ORL	118	14	0.119	0.737	0.250	1.000	0.279
clinic	gynecology	274	24	0.088	0.805	0.425	0.750	0.668
clinic	neuro	488	25	0.051	0.852 (0.752–0.930)	0.512 (0.317–0.692)	0.840 (0.682–0.963)	0.678 (0.635–0.719)
clinic	ortho	971	76	0.078	0.822 (0.778–0.862)	0.282 (0.206–0.387)	0.908 (0.837–0.967)	0.581 (0.548–0.613)
clinic	pediatric	37	3	0.081	0.652	0.227	0.667	0.412
clinic	thoracic	403	14	0.035	0.936	0.479	0.929	0.753
clinic	urology	200	19	0.095	0.762	0.374	0.737	0.564
clinic	vascular	2838	183	0.064	0.844 (0.809–0.878)	0.371 (0.298–0.444)	0.885 (0.837–0.932)	0.591 (0.573–0.610)
clinic	visceral	722	143	0.198	0.862 (0.831–0.893)	0.642 (0.562–0.719)	0.965 (0.934–0.993)	0.454 (0.416–0.496)
surgical_group	Overall	6051	501	0.083	0.855 (0.836–0.873)	0.440 (0.394–0.486)	0.900 (0.873–0.926)	0.590 (0.576–0.602)
surgical_group	chopcluster_1	395	21	0.053	0.868	0.525	0.857	0.679
surgical_group	chopcluster_10	186	28	0.151	0.775 (0.655–0.883)	0.474 (0.308–0.692)	0.786 (0.625–0.929)	0.500 (0.422–0.574)
surgical_group	chopcluster_11	178	9	0.051	0.739	0.260	0.556	0.751
surgical_group	chopcluster_12	826	61	0.074	0.825 (0.773–0.872)	0.259 (0.182–0.380)	0.918 (0.842–0.981)	0.603 (0.567–0.639)
surgical_group	chopcluster_13	154	43	0.279	0.752 (0.664–0.833)	0.584 (0.446–0.722)	0.930 (0.846–1.000)	0.216 (0.142–0.292)
surgical_group	chopcluster_3	414	22	0.053	0.932	0.547	0.955	0.730
surgical_group	chopcluster_4	2245	118	0.053	0.819 (0.767–0.865)	0.299 (0.232–0.391)	0.839 (0.770–0.903)	0.613 (0.593–0.633)

Attribute	Subgroup	n	Events	Prevalence	AUROC (95% CI)	AUPRC (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)
surgical_group	chopcluster_5	328	30	0.091	0.856 (0.781–0.920)	0.465 (0.297–0.642)	0.933 (0.833–1.000)	0.527 (0.472–0.581)
surgical_group	chopcluster_6	373	66	0.177	0.877 (0.829–0.920)	0.691 (0.583–0.784)	0.955 (0.900–1.000)	0.463 (0.407–0.518)
surgical_group	chopcluster_7	143	36	0.252	0.826 (0.755–0.890)	0.634 (0.479–0.777)	1.000 (1.000–1.000)	0.271 (0.192–0.357)
surgical_group	chopcluster_8	142	23	0.162	0.785	0.439	0.870	0.513
surgical_group	none	667	44	0.066	0.880 (0.836–0.919)	0.395 (0.259–0.546)	0.977 (0.921–1.000)	0.559 (0.521–0.599)
emergency	Overall	6051	501	0.083	0.855 (0.836–0.873)	0.440 (0.394–0.486)	0.900 (0.873–0.926)	0.590 (0.576–0.602)
emergency	elective	5162	314	0.061	0.835 (0.810–0.859)	0.303 (0.260–0.360)	0.847 (0.807–0.887)	0.645 (0.631–0.658)
emergency	emergency	889	187	0.210	0.830 (0.796–0.863)	0.626 (0.557–0.695)	0.989 (0.972–1.000)	0.207 (0.178–0.237)
sex × age_group	Overall	6051	501	0.083	0.855 (0.836–0.873)	0.440 (0.394–0.486)	0.900 (0.873–0.926)	0.590 (0.576–0.602)
sex × age_group	female / 0-17	109	13	0.101	0.893	0.491	0.911	0.525
sex × age_group	female / 18-44	220	13	0.073	0.840	0.363	0.901	0.600
sex × age_group	female / 45-64	683	52	0.076	0.838 (0.764–0.899)	0.413 (0.292–0.553)	0.846 (0.738–0.939)	0.624 (0.586–0.660)
sex × age_group	female / 65-79	839	92	0.110	0.885 (0.848–0.920)	0.601 (0.502–0.694)	0.946 (0.892–0.989)	0.592 (0.557–0.628)
sex × age_group	female / 80+	239	19	0.084	0.773	0.220	0.899	0.532
sex × age_group	male / 0-17	114	8	0.075	0.647	0.175	0.886	0.543
sex × age_group	male / 18-44	298	21	0.074	0.811	0.438	0.888	0.612
sex × age_group	male / 45-64	1366	94	0.069	0.883 (0.850–0.915)	0.448 (0.351–0.551)	0.957 (0.910–0.990)	0.603 (0.575–0.629)
sex × age_group	male / 65-79	1859	155	0.083	0.863 (0.827–0.895)	0.464 (0.387–0.545)	0.897 (0.848–0.942)	0.601 (0.577–0.624)
sex × age_group	male / 80+	324	34	0.096	0.765	0.335	0.890	0.541
sex × frailty	Overall	6051	501	0.083	0.855 (0.836–0.873)	0.440 (0.394–0.486)	0.900 (0.873–0.926)	0.590 (0.576–0.602)
sex × frailty	female / frail	1010	129	0.128	0.869 (0.833–0.902)	0.540 (0.457–0.624)	0.938 (0.893–0.977)	0.491 (0.460–0.526)
sex × frailty	female / non_frail	275	1	0.041	0.993	0.333	0.905	0.684
sex × frailty	female / pre_frail	805	59	0.073	0.791 (0.721–0.850)	0.277 (0.183–0.405)	0.831 (0.729–0.921)	0.630 (0.595–0.662)
sex × frailty	male / frail	2674	270	0.101	0.854 (0.830–0.878)	0.472 (0.410–0.537)	0.922 (0.890–0.953)	0.549 (0.529–0.567)
sex × frailty	male / non_frail	272	5	0.045	0.825	0.120	0.895	0.712
sex × frailty	male / pre_frail	1015	37	0.044	0.785	0.218	0.872	0.633
age_group × frailty	Overall	6051	501	0.083	0.855 (0.836–0.873)	0.440 (0.394–0.486)	0.900 (0.873–0.926)	0.590 (0.576–0.602)
age_group × frailty	0-17 / frail	128	19	0.121	0.779	0.398	0.862	0.379
age_group × frailty	0-17 / non_frail	28	0	0.078	—	—	—	0.714
age_group × frailty	0-17 / pre_frail	67	2	0.073	0.538	0.061	0.852	0.478
age_group × frailty	18-44 / frail	183	21	0.094	0.836	0.549	0.835	0.517
age_group × frailty	18-44 / non_frail	182	2	0.034	0.819	0.042	0.826	0.717
age_group × frailty	18-44 / pre_frail	153	11	0.069	0.706	0.186	0.809	0.620

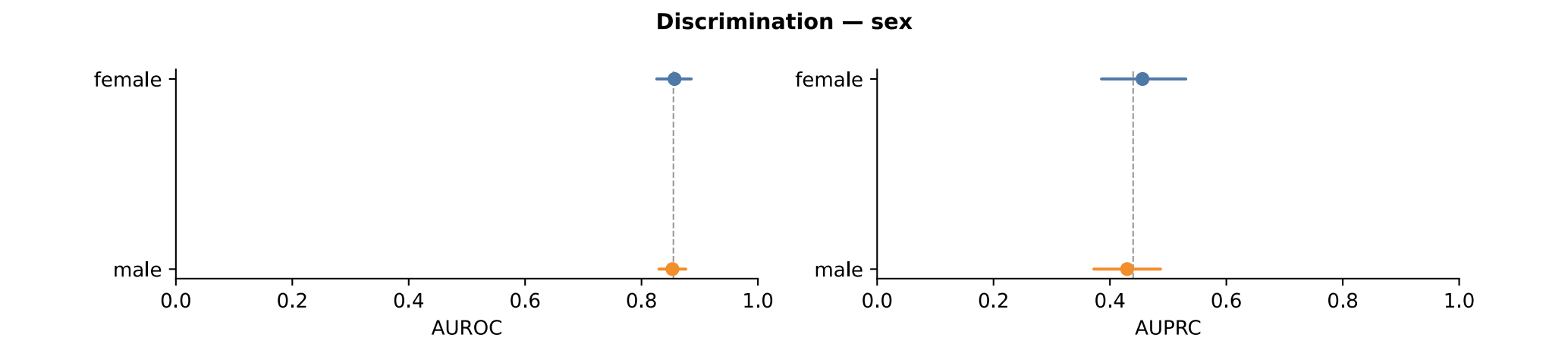
Attribute	Subgroup	n	Events	Prevalence	AUROC (95% CI)	AUPRC (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)
age_group × frailty	45-64 / frail	1167	110	0.094	0.880 (0.848–0.912)	0.489 (0.399–0.588)	0.964 (0.927–0.993)	0.556 (0.526–0.586)
age_group × frailty	45-64 / non_frail	210	3	0.036	0.960	0.373	0.920	0.730
age_group × frailty	45-64 / pre_frail	672	33	0.053	0.775	0.222	0.886	0.631
age_group × frailty	65-79 / frail	1846	204	0.111	0.874 (0.844–0.903)	0.566 (0.495–0.638)	0.922 (0.883–0.958)	0.563 (0.540–0.587)
age_group × frailty	65-79 / non_frail	118	1	0.053	0.517	0.017	0.908	0.678
age_group × frailty	65-79 / pre_frail	734	42	0.063	0.847	0.308	0.913	0.644
age_group × frailty	80+ / frail	360	45	0.117	0.751	0.318	0.859	0.455
age_group × frailty	80+ / non_frail	9	0	0.088	—	—	—	0.444
age_group × frailty	80+ / pre_frail	194	8	0.063	0.714	0.244	0.836	0.570

Gap statistics

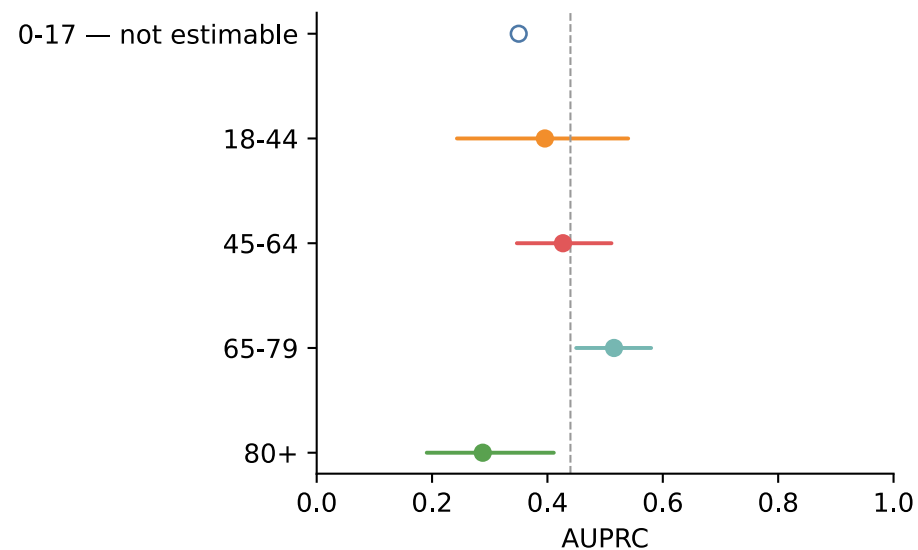
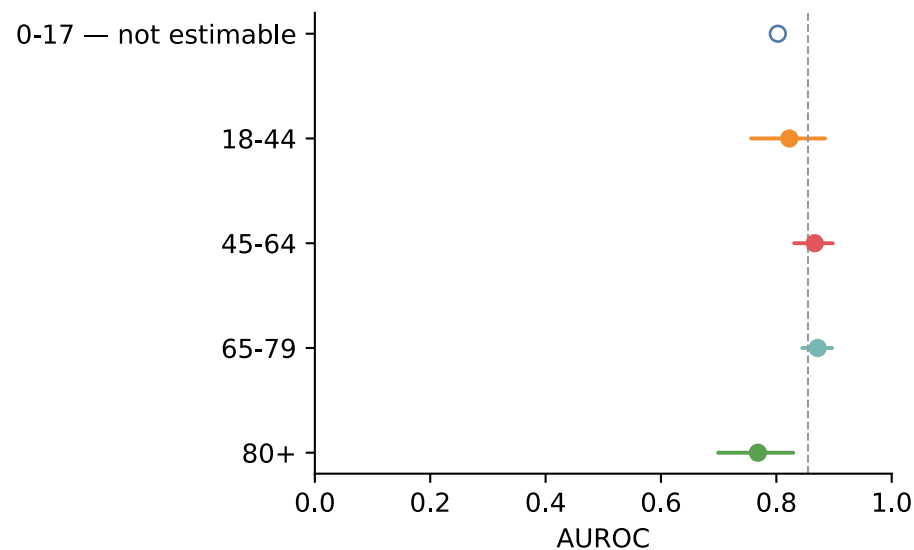
Effect-size column shows Cohen's *h* on rate-metric gaps for two-group attributes; multi-group attributes and ranking metrics (AUROC / AUPRC) show "—".

Attribute	Metric	Value	Effect size
sex	equal_opportunity_difference	0.0073	h=+0.02 (negligible)
sex	predictive_equality	0.0078	h=-0.02 (negligible)
sex	equalized_odds_gap	0.0078	—
sex	statistical_parity	0.0135	—
age_group	equal_opportunity_difference	0.0943	—
age_group	predictive_equality	0.1792	—
age_group	equalized_odds_gap	0.1792	—
age_group	statistical_parity	0.1782	—
frailty	equal_opportunity_difference	0.1357	—
frailty	predictive_equality	0.2486	—
frailty	equalized_odds_gap	0.2486	—
frailty	statistical_parity	0.2917	—
clinic	equal_opportunity_difference	0.3333	—
clinic	predictive_equality	0.4744	—
clinic	equalized_odds_gap	0.4744	—
clinic	statistical_parity	0.4838	—
surgical_group	equal_opportunity_difference	0.4444	—
surgical_group	predictive_equality	0.5353	—
surgical_group	equalized_odds_gap	0.5353	—

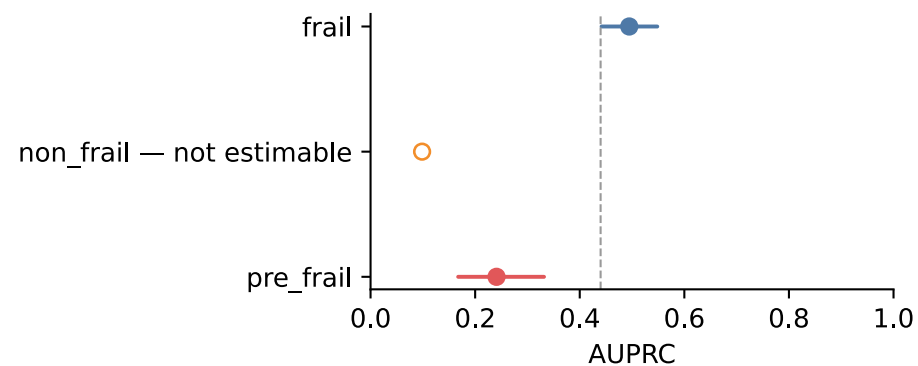
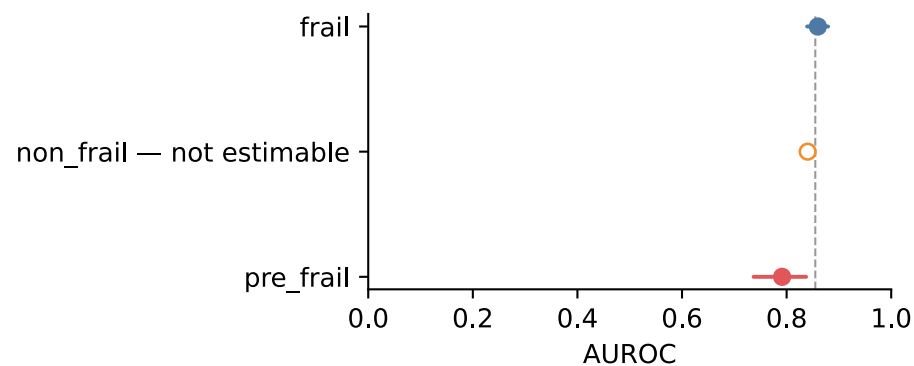
Attribute	Metric	Value	Effect size
surgical_group	statistical_parity	0.5606	—
emergency	equal_opportunity_difference	0.1422	h=−0.60 (medium)
emergency	predictive_equality	0.4385	h=+0.92 (large)
emergency	equalized_odds_gap	0.4385	—
emergency	statistical_parity	0.4497	—
sex × age_group	equal_opportunity_difference	0.1113	—
sex × age_group	predictive_equality	0.0993	—
sex × age_group	equalized_odds_gap	0.1113	—
sex × age_group	statistical_parity	0.1156	—
sex × frailty	equal_opportunity_difference	0.1075	—
sex × frailty	predictive_equality	0.2203	—
sex × frailty	equalized_odds_gap	0.2203	—
sex × frailty	statistical_parity	0.2640	—
age_group × frailty	equal_opportunity_difference	0.1543	—
age_group × frailty	predictive_equality	0.3510	—
age_group × frailty	equalized_odds_gap	0.3510	—
age_group × frailty	statistical_parity	0.3774	—



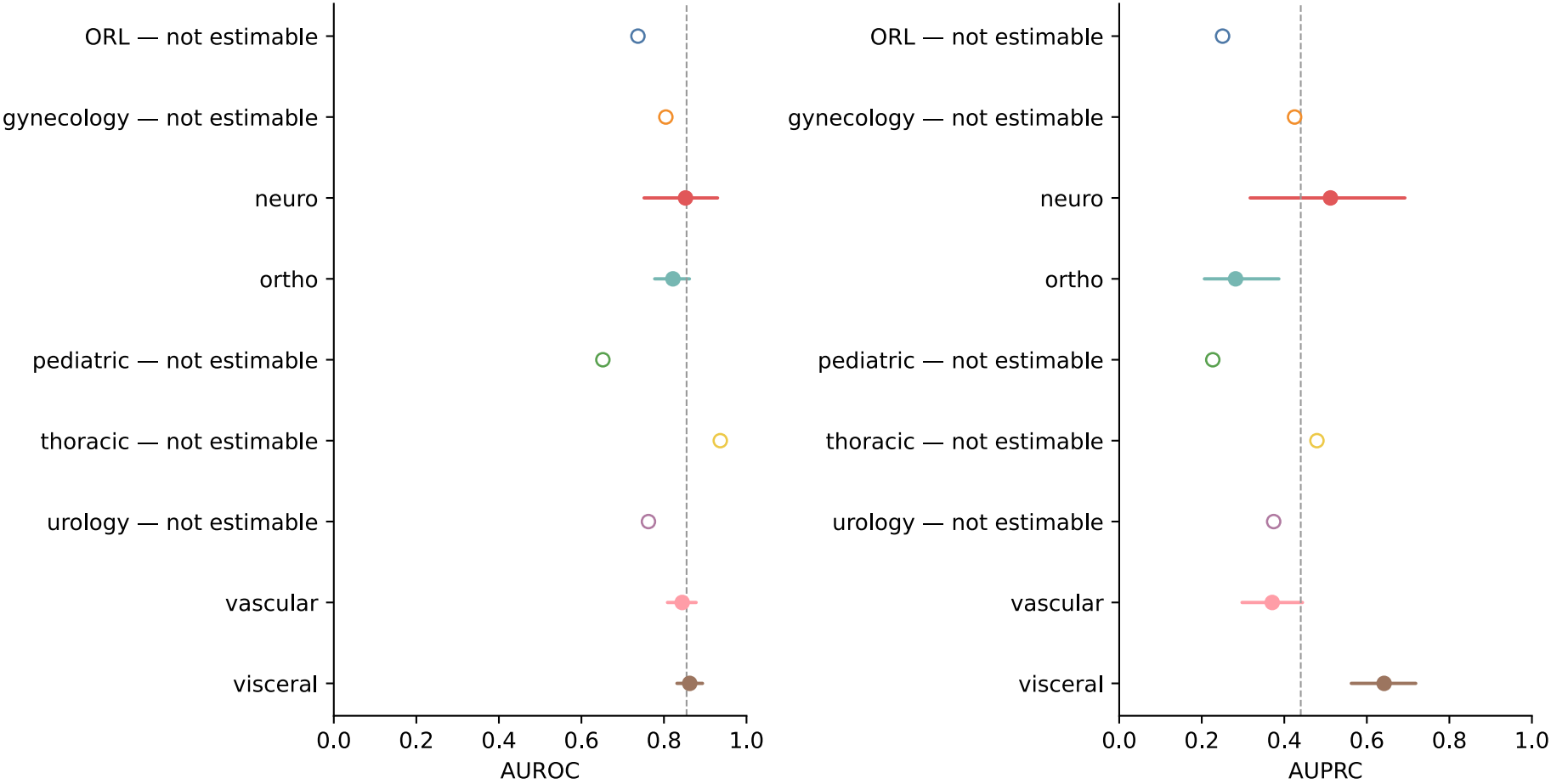
Discrimination — age_group



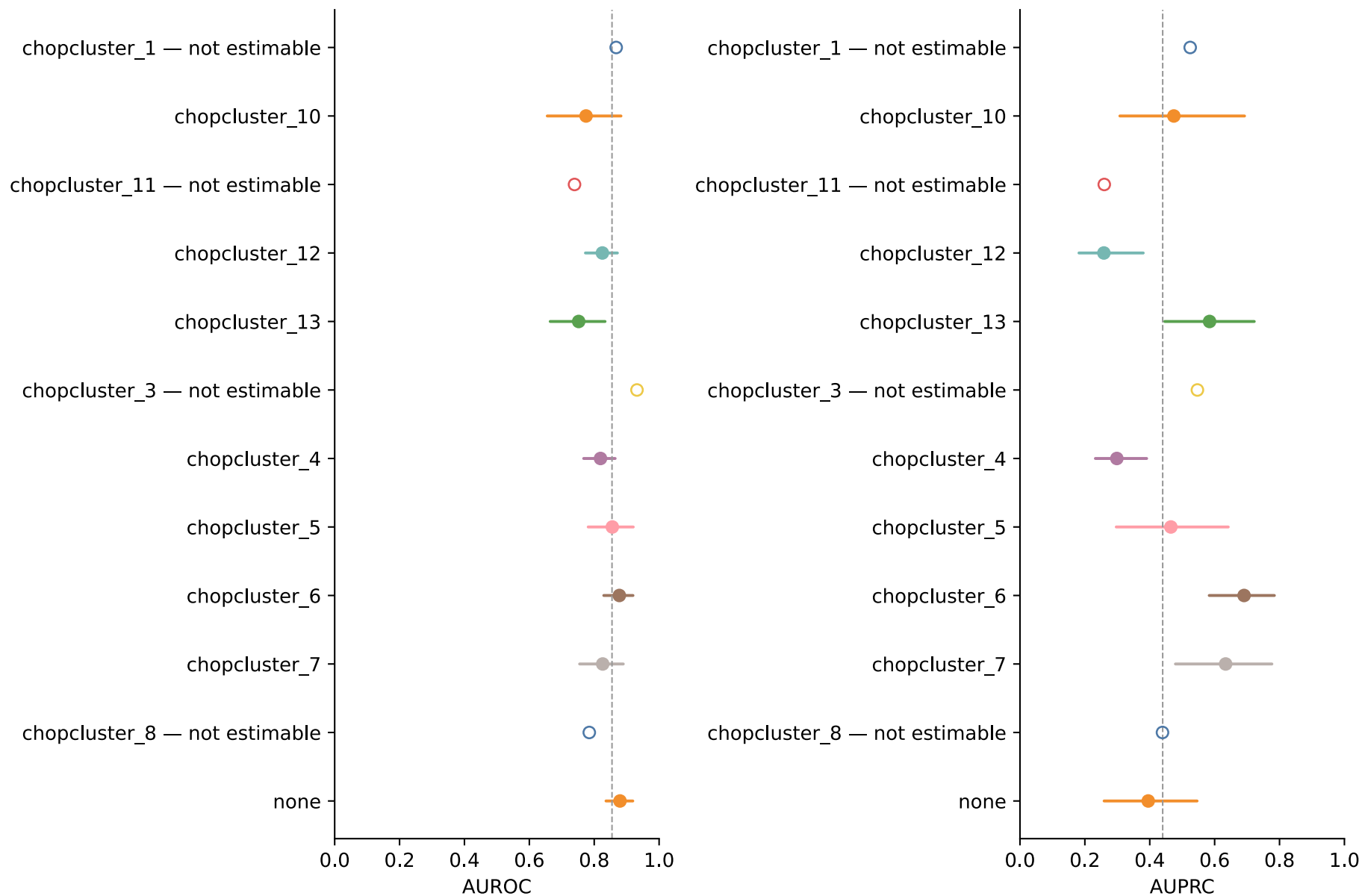
Discrimination — frailty

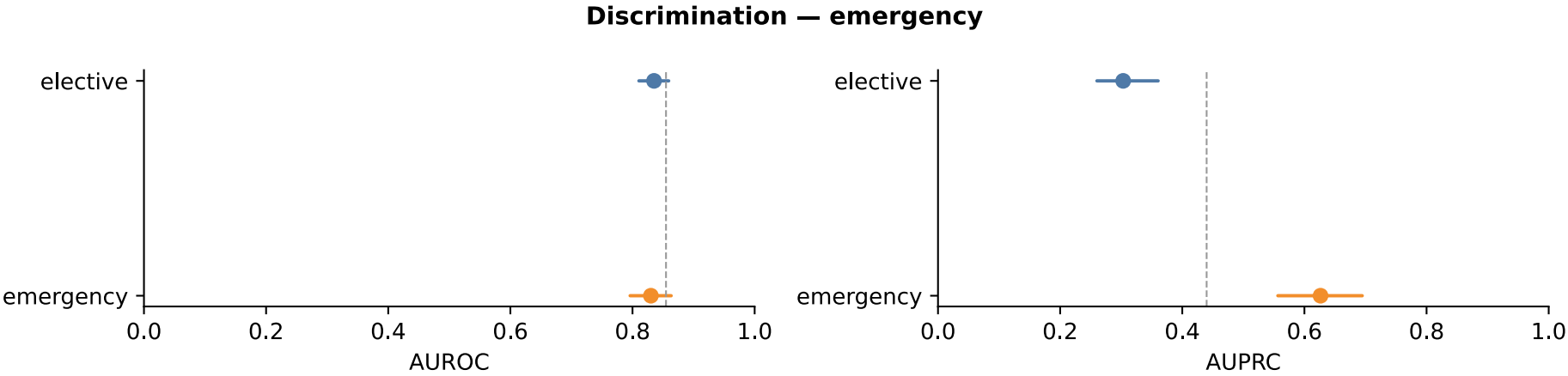


Discrimination — clinic

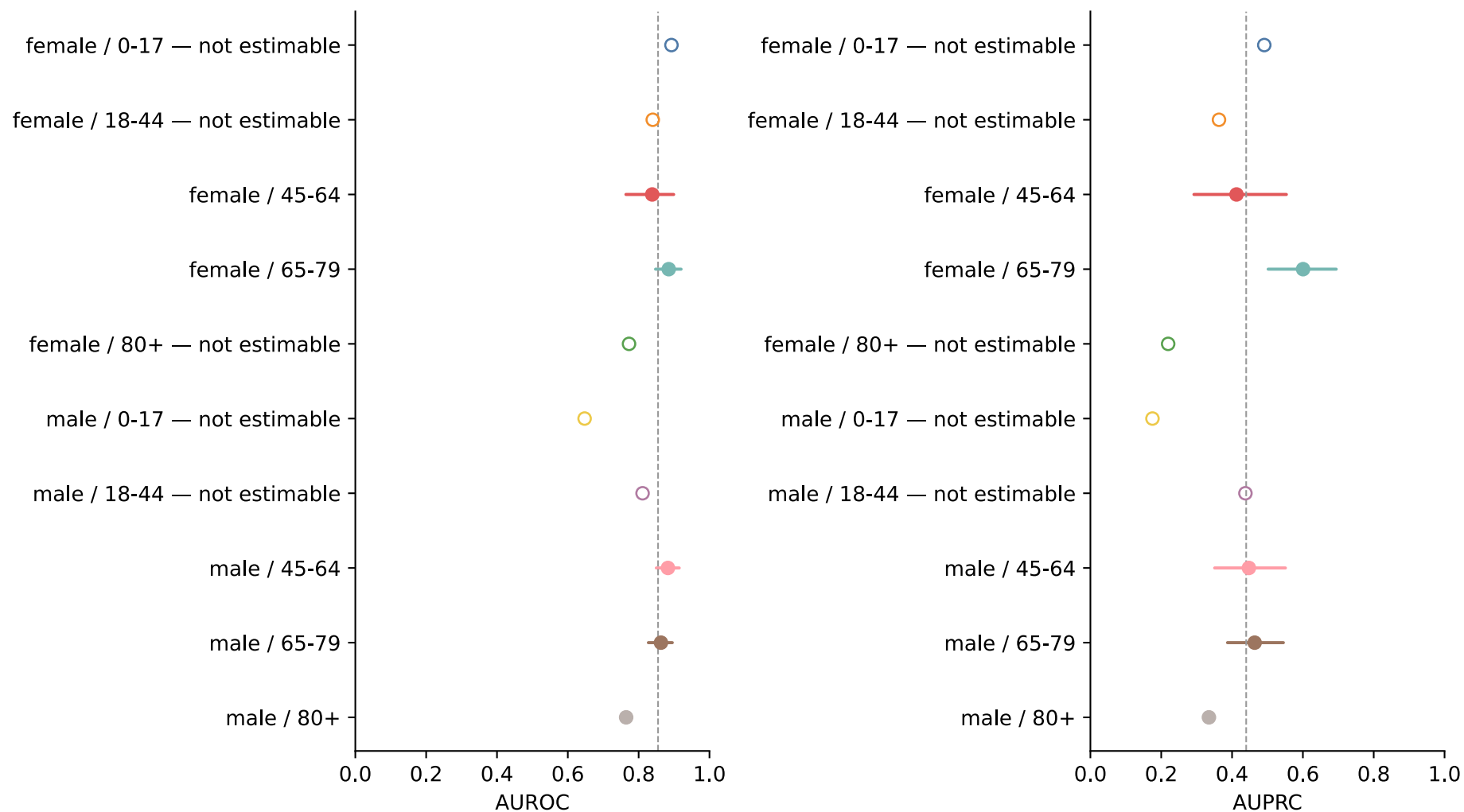


Discrimination — surgical_group

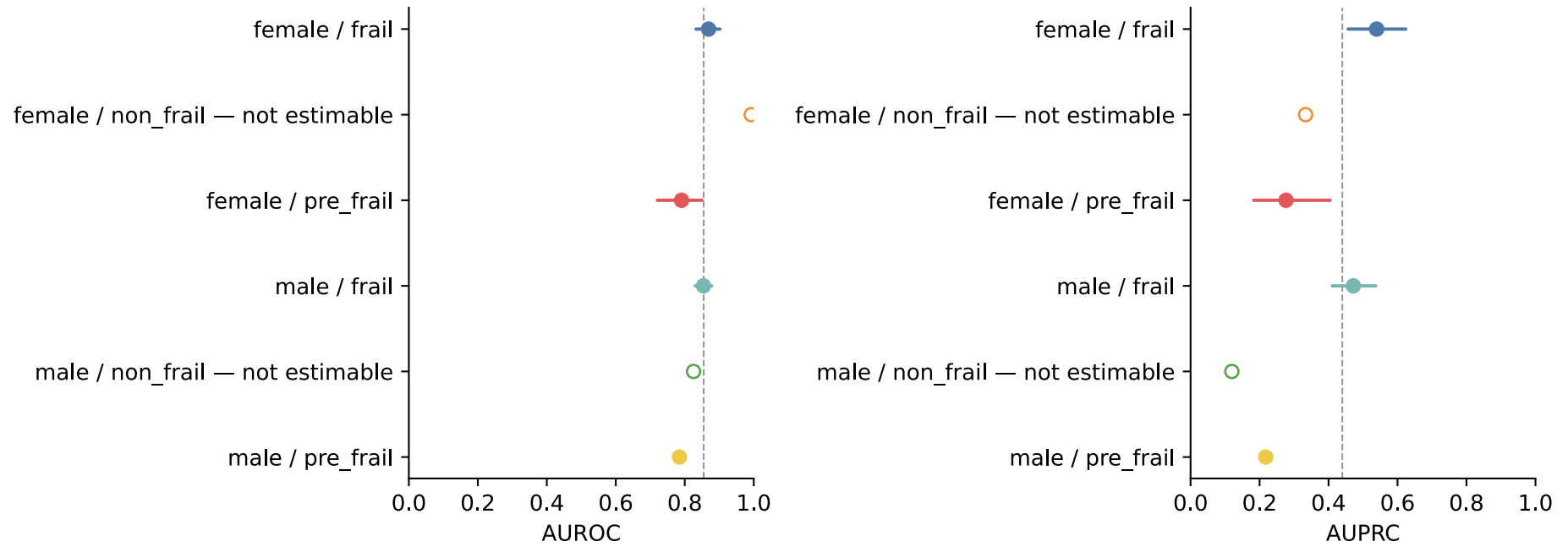




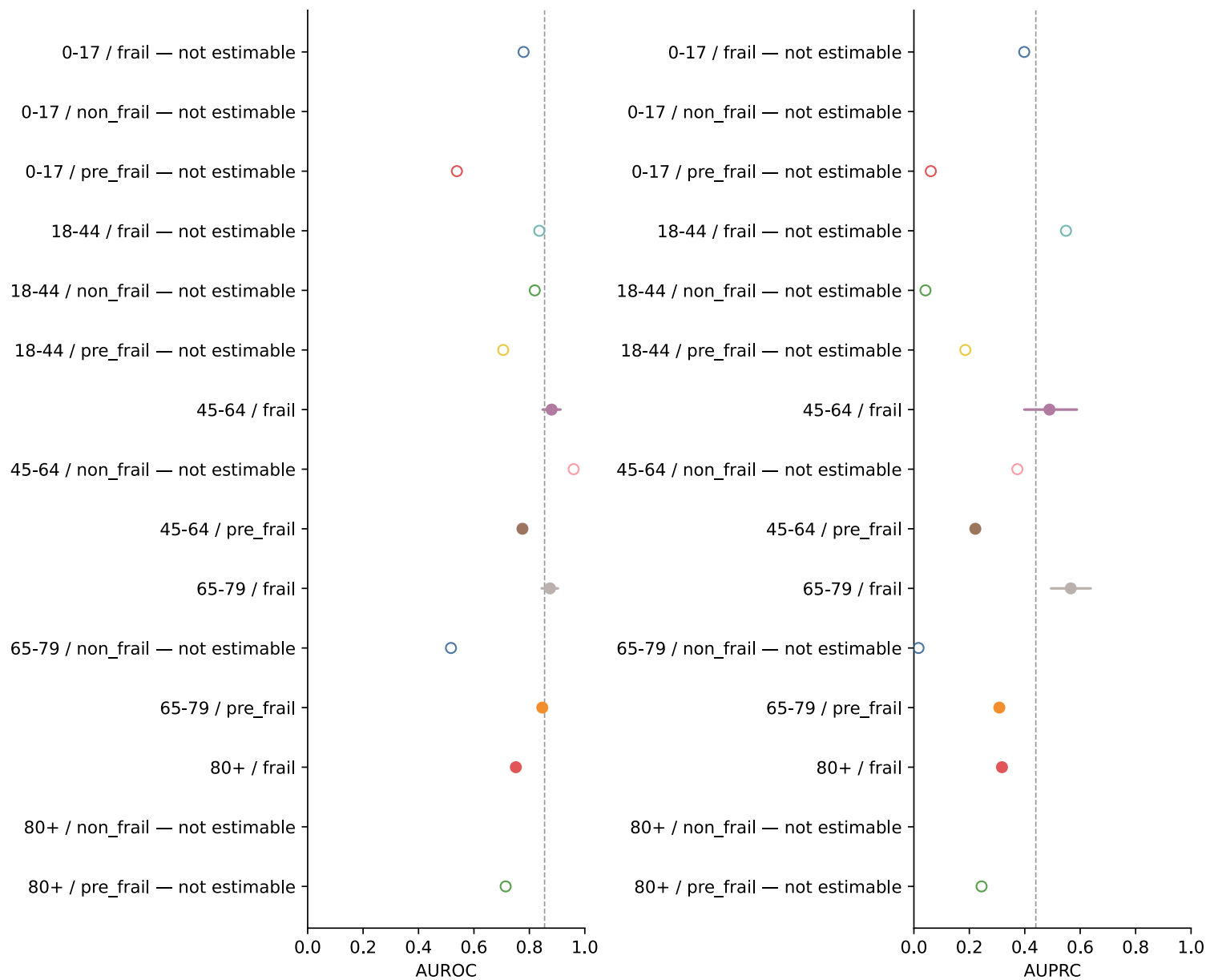
Discrimination — sex × age_group



Discrimination — sex × frailty

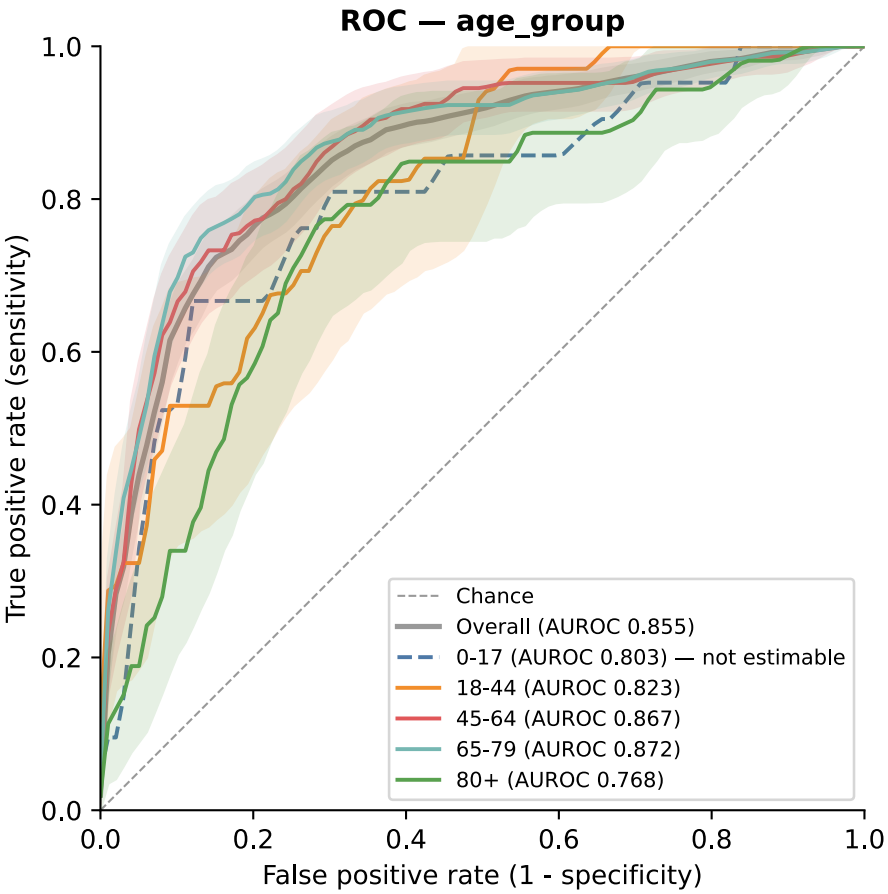
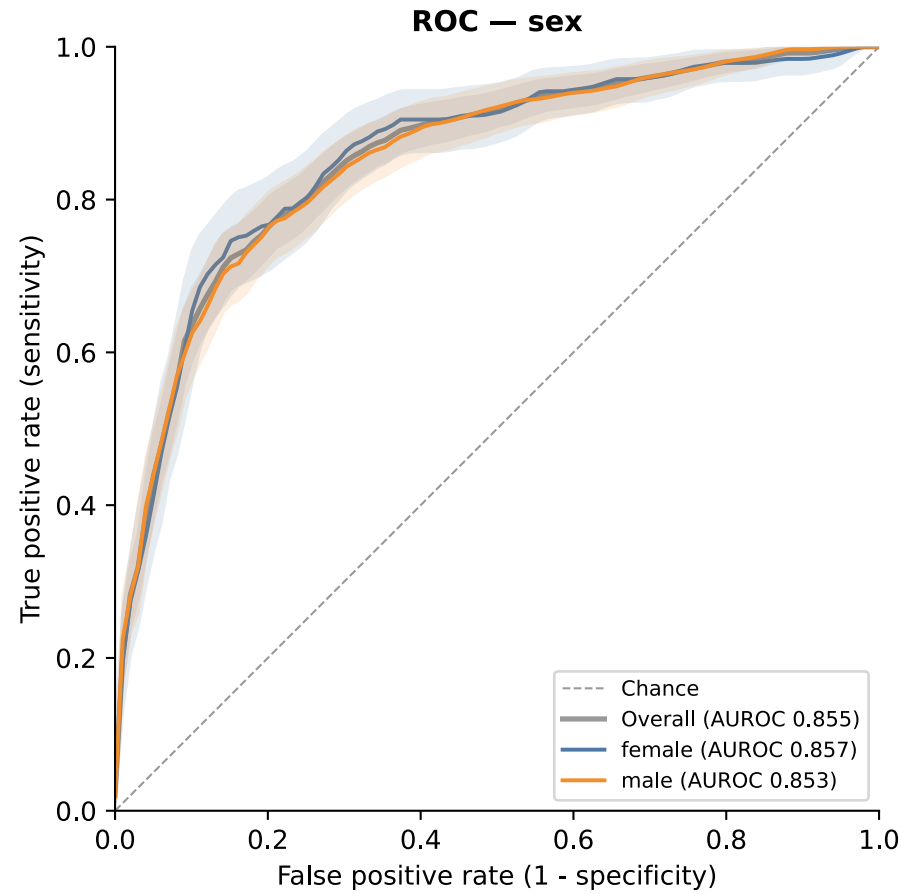


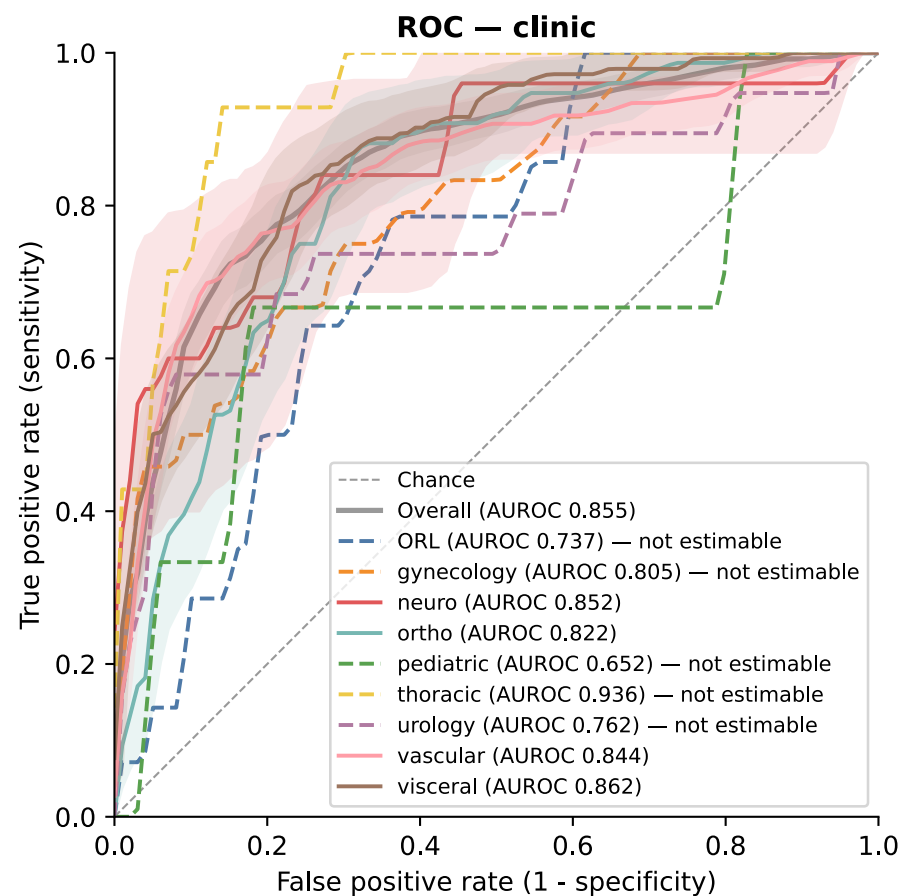
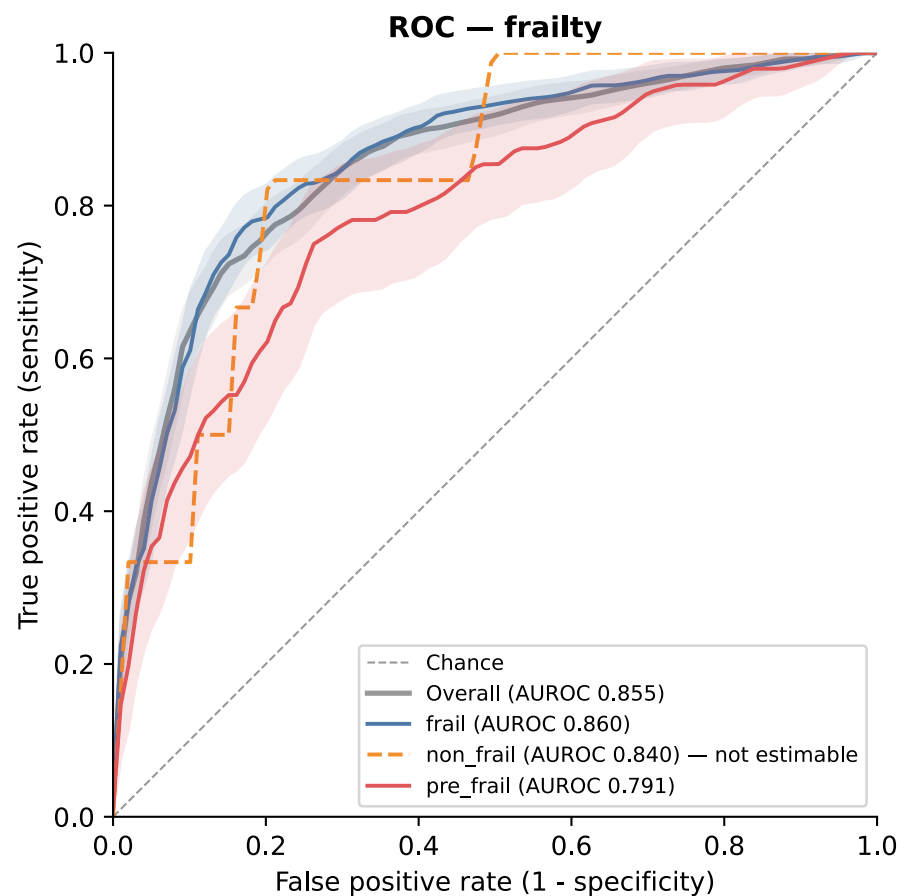
Discrimination — age_group × frailty

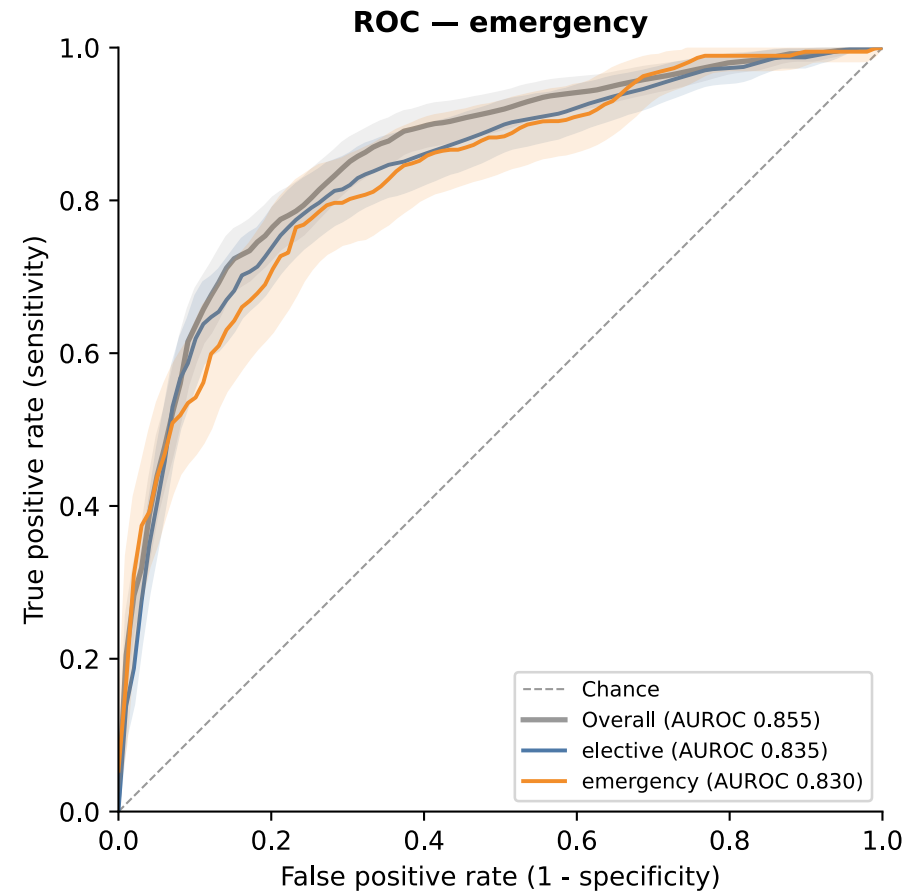
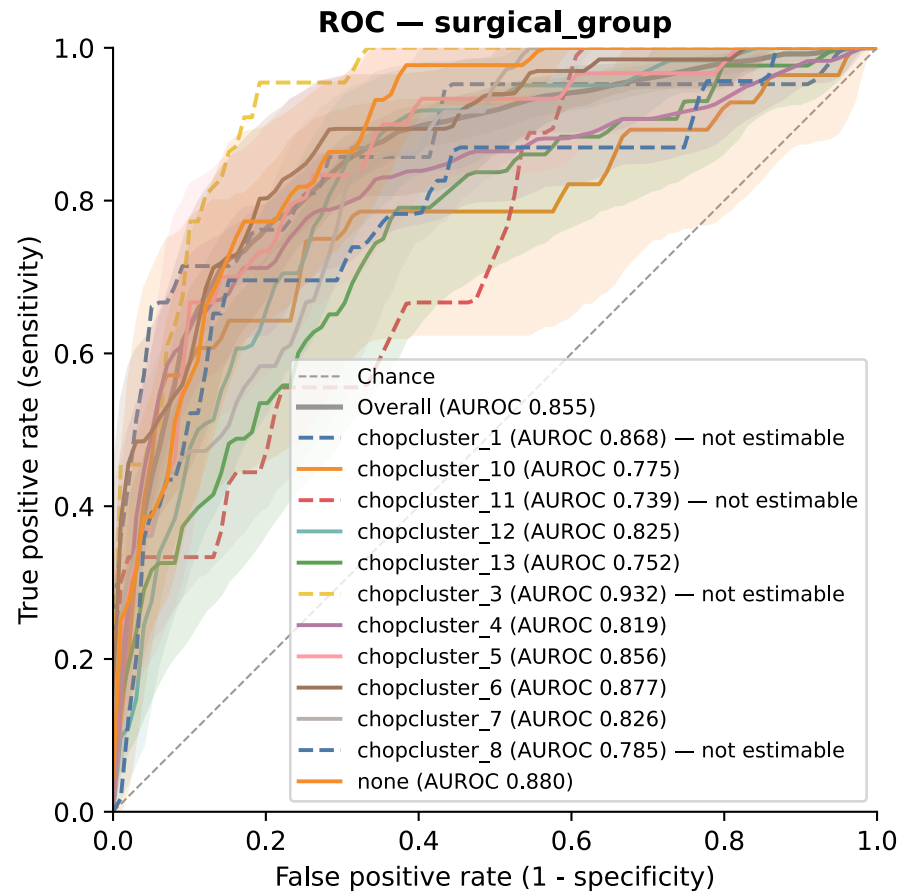


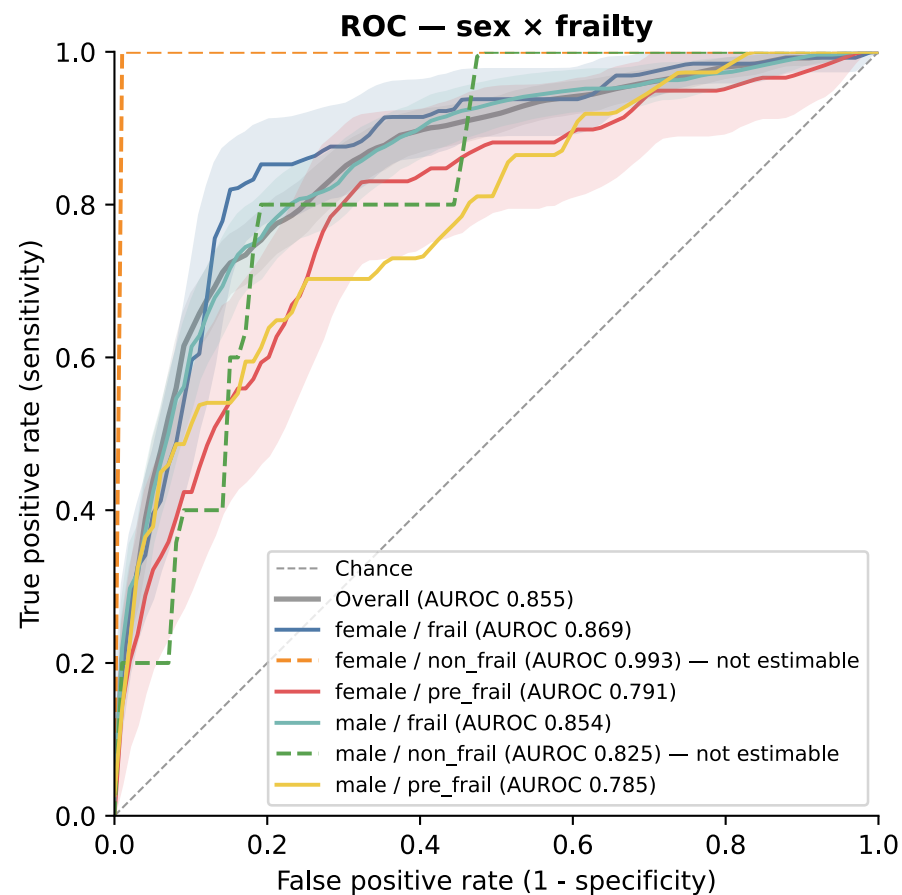
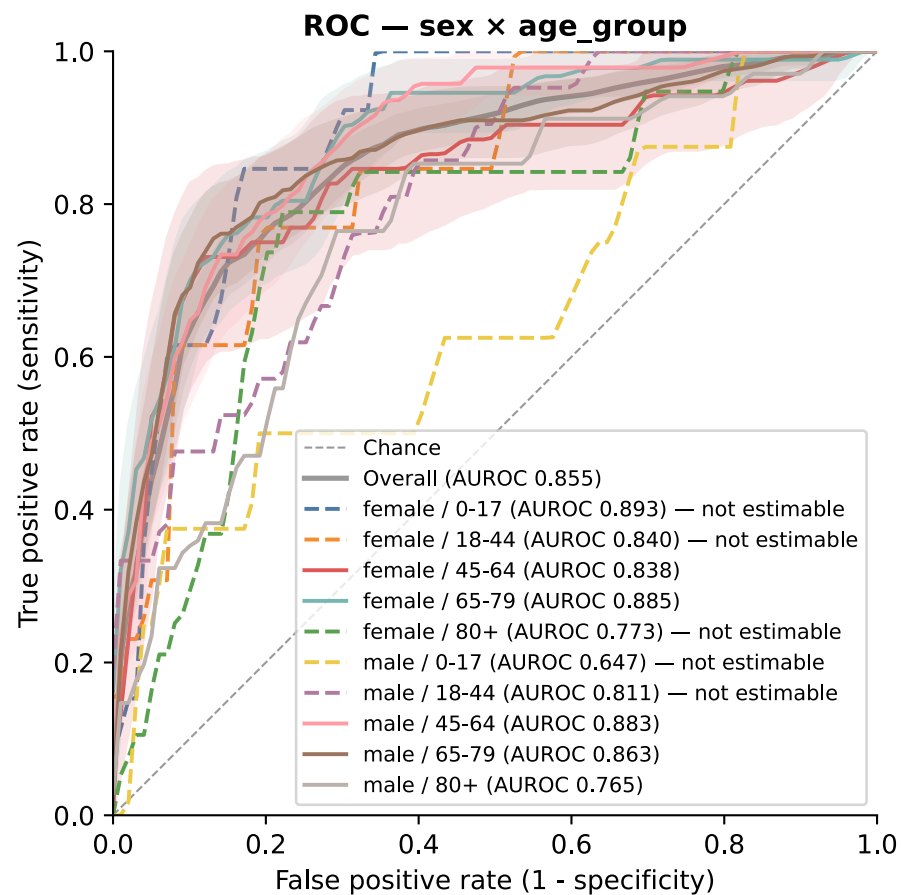
ROC curves

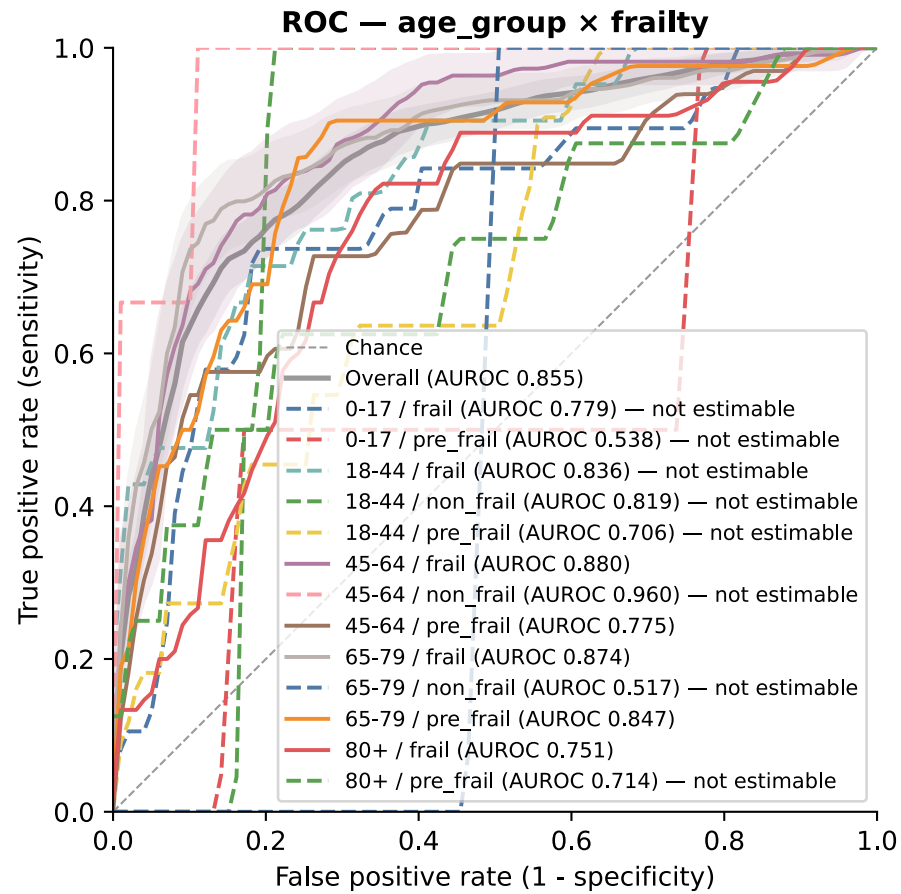
Per-subgroup ROC curves with pointwise 95% bootstrap bands. AUROC values in the legend are taken from the table above, not integrated from the drawn curve. Curves that cross indicate that no single subgroup dominates at every operating point — the ranking of subgroups depends on the threshold chosen.











Calibration

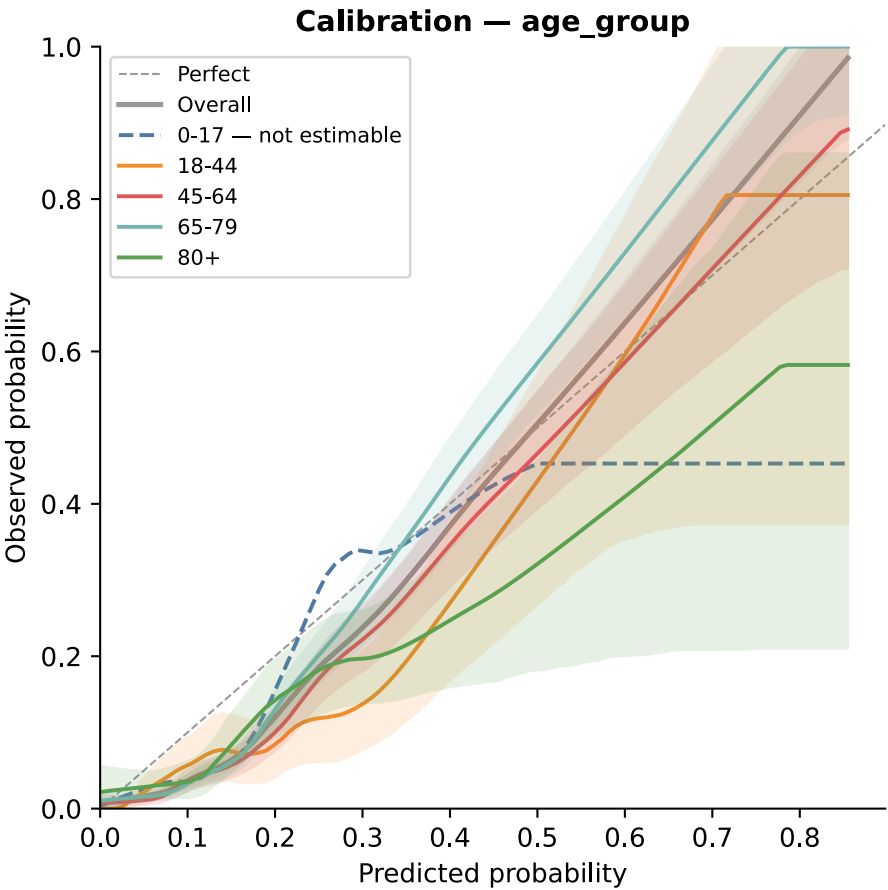
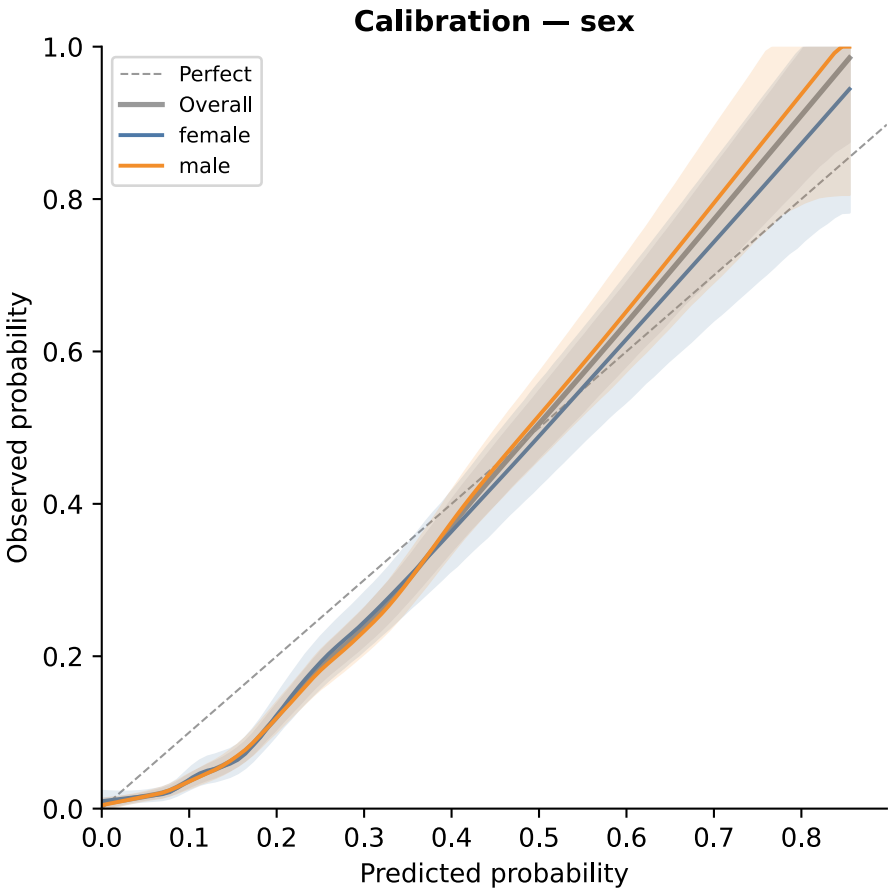
Attribute	Subgroup	n	Intercept (95% CI)	Slope (95% CI)	Brier (95% CI)	ECE	ICI
sex	Overall	6051	-0.579 (-0.673–0.489)	1.441 (1.325–1.578)	0.062 (0.058–0.066)	0.0472	0.0470
sex	female	2090	-0.540 (-0.698–0.390)	1.395 (1.210–1.613)	0.066 (0.059–0.073)	0.0465	0.0453
sex	male	3961	-0.601 (-0.719–0.491)	1.469 (1.316–1.641)	0.060 (0.055–0.065)	0.0479	0.0482
age_group	Overall	6051	-0.579 (-0.673–0.489)	1.441 (1.325–1.578)	0.062 (0.058–0.066)	0.0472	0.0470
age_group	0-17	223	-0.463	1.470	0.076	0.0519	0.0469
age_group	18-44	518	-0.762 (-1.183–0.441)	1.240	0.056 (0.044–0.069)	0.0574	0.0547
age_group	45-64	2049	-0.710 (-0.884–0.555)	1.434 (1.221–1.693)	0.055 (0.049–0.061)	0.0532	0.0514
age_group	65-79	2698	-0.467 (-0.601–0.344)	1.597 (1.411–1.822)	0.063 (0.057–0.069)	0.0462	0.0478
age_group	80+	563	-0.604 (-0.914–0.315)	0.997	0.082 (0.068–0.097)	0.0567	0.0585
frailty	Overall	6051	-0.579 (-0.673–0.489)	1.441 (1.325–1.578)	0.062 (0.058–0.066)	0.0472	0.0470
frailty	frail	3684	-0.438 (-0.551–0.340)	1.490 (1.349–1.645)	0.075 (0.069–0.080)	0.0441	0.0447
frailty	non_frail	547	-1.996	1.357	0.017	0.0589	0.0590
frailty	pre_frail	1820	-0.843 (-1.068–0.653)	1.112	0.050 (0.043–0.057)	0.0548	0.0536
clinic	Overall	6051	-0.579 (-0.673–0.489)	1.441 (1.325–1.578)	0.062 (0.058–0.066)	0.0472	0.0470
clinic	ORL	118	-0.776	0.950	0.110	0.1037	0.0969
clinic	gynecology	274	-0.262	1.225	0.066	0.0315	0.0276
clinic	neuro	488	-0.720 (-1.196–0.370)	1.755	0.040 (0.029–0.053)	0.0474	0.0528
clinic	ortho	971	-0.635 (-0.889–0.413)	1.166	0.067 (0.058–0.078)	0.0511	0.0513
clinic	pediatric	37	-0.675	0.572	0.080	0.1231	0.0780
clinic	thoracic	403	-1.072	1.891	0.029	0.0514	0.0504
clinic	urology	200	-0.302	1.115	0.074	0.0692	0.0409
clinic	vascular	2838	-0.740 (-0.898–0.601)	1.539 (1.315–1.803)	0.053 (0.048–0.058)	0.0544	0.0533
clinic	visceral	722	-0.166 (-0.362–0.028)	1.432 (1.215–1.754)	0.111 (0.099–0.124)	0.0516	0.0431
surgical_group	Overall	6051	-0.579 (-0.673–0.489)	1.441 (1.325–1.578)	0.062 (0.058–0.066)	0.0472	0.0470
surgical_group	chopcluster_1	395	-0.759	1.692	0.041	0.0516	0.0534
surgical_group	chopcluster_10	186	0.106 (-0.355–0.483)	1.205	0.105 (0.074–0.135)	0.0946	0.0568
surgical_group	chopcluster_11	178	-0.676	0.953	0.047	0.0422	0.0401
surgical_group	chopcluster_12	826	-0.627 (-0.908–0.376)	1.181	0.064 (0.054–0.075)	0.0484	0.0477
surgical_group	chopcluster_13	154	0.179 (-0.214–0.534)	1.114	0.167 (0.135–0.200)	0.0566	0.0315
surgical_group	chopcluster_3	414	-0.807	1.756	0.039	0.0477	0.0486
surgical_group	chopcluster_4	2245	-0.869 (-1.067–0.694)	1.463 (1.196–1.792)	0.047 (0.041–0.052)	0.0577	0.0562

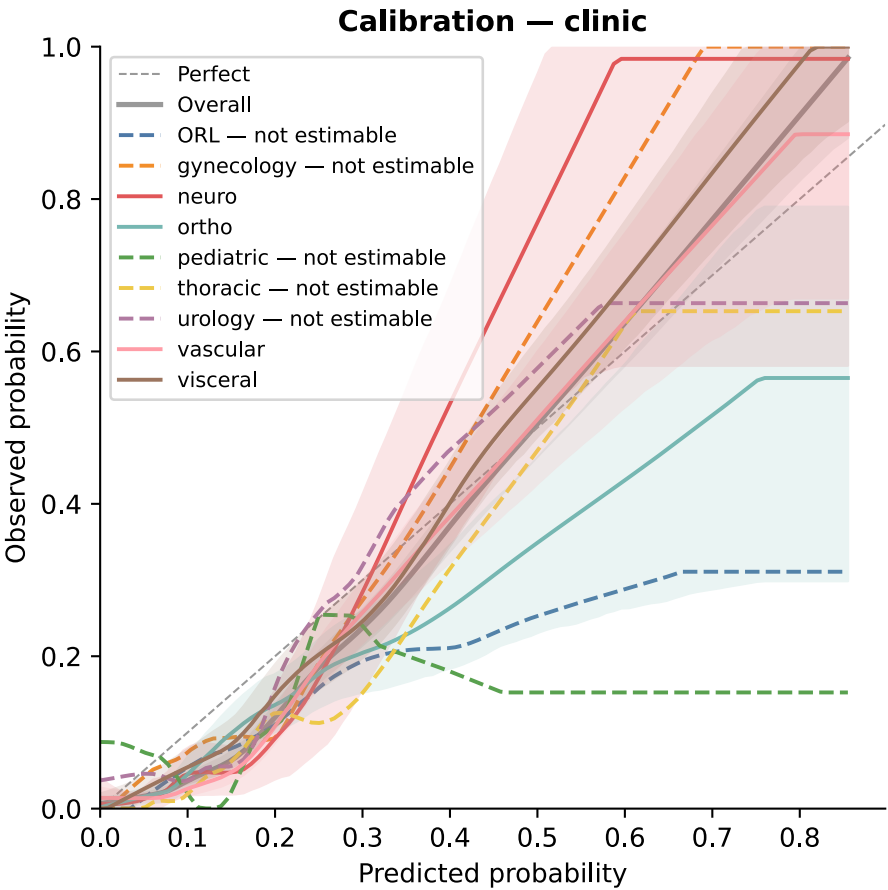
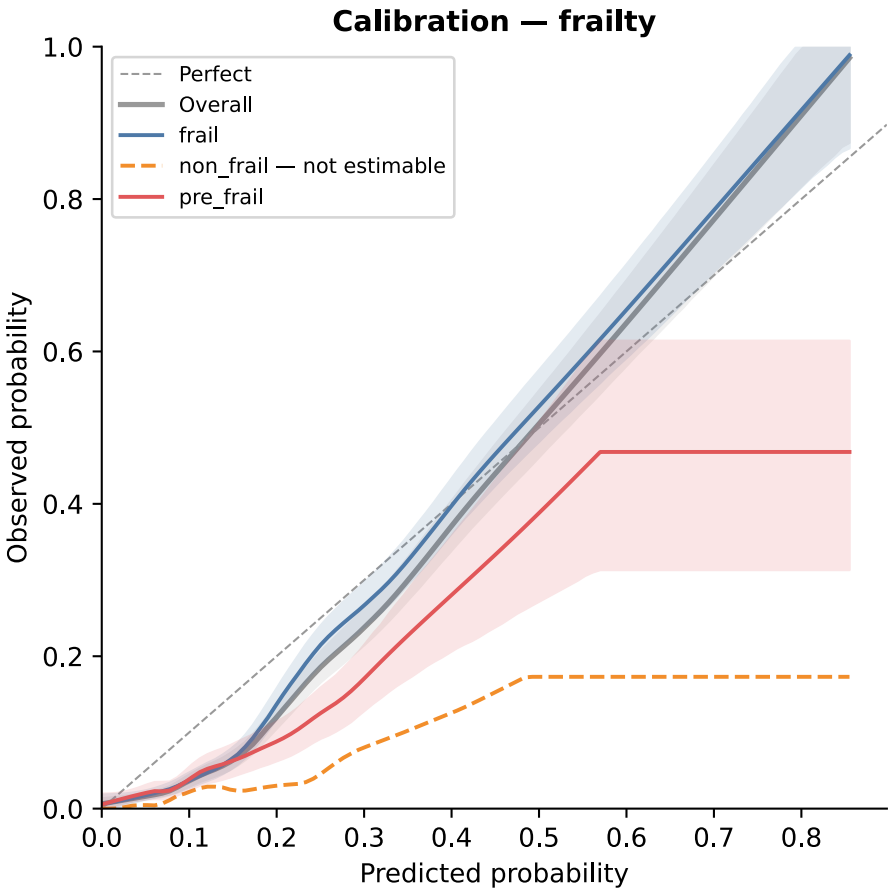
Attribute	Subgroup	n	Intercept (95% CI)	Slope (95% CI)	Brier (95% CI)	ECE	ICI
surgical_group	chopcluster_5	328	-0.574 (-0.998–0.236)	1.510	0.068 (0.052–0.085)	0.0555	0.0511
surgical_group	chopcluster_6	373	-0.245 (-0.525–0.011)	1.616	0.098 (0.082–0.115)	0.0717	0.0571
surgical_group	chopcluster_7	143	-0.082 (-0.478–0.267)	1.517	0.141 (0.113–0.171)	0.0784	0.0405
surgical_group	chopcluster_8	142	-0.157	0.909	0.112	0.0572	0.0288
surgical_group	none	667	-0.893 (-1.249–0.604)	1.508	0.056 (0.046–0.066)	0.0677	0.0661
emergency	Overall	6051	-0.579 (-0.673–0.489)	1.441 (1.325–1.578)	0.062 (0.058–0.066)	0.0472	0.0470
emergency	elective	5162	-0.709 (-0.823–0.592)	1.359 (1.210–1.517)	0.051 (0.048–0.056)	0.0485	0.0473
emergency	emergency	889	-0.263 (-0.436–0.102)	1.577 (1.338–1.862)	0.123 (0.112–0.135)	0.0724	0.0640
sex × age_group	Overall	6051	-0.579 (-0.673–0.489)	1.441 (1.325–1.578)	0.062 (0.058–0.066)	0.0472	0.0470
sex × age_group	female / 0-17	109	-0.317	2.089	0.085	0.0679	0.0682
sex × age_group	female / 18-44	220	-0.959	1.284	0.054	0.0723	0.0678
sex × age_group	female / 45-64	683	-0.668 (-1.006–0.395)	1.183	0.059 (0.049–0.071)	0.0544	0.0504
sex × age_group	female / 65-79	839	-0.328 (-0.549–0.122)	1.692	0.069 (0.059–0.080)	0.0469	0.0467
sex × age_group	female / 80+	239	-0.830	0.965	0.076	0.0758	0.0775
sex × age_group	male / 0-17	114	-0.656	0.786	0.068	0.0783	0.0531
sex × age_group	male / 18-44	298	-0.622	1.235	0.058	0.0465	0.0476
sex × age_group	male / 45-64	1366	-0.732 (-0.950–0.542)	1.613	0.053 (0.046–0.060)	0.0544	0.0536
sex × age_group	male / 65-79	1859	-0.539 (-0.714–0.382)	1.544 (1.310–1.808)	0.060 (0.053–0.067)	0.0486	0.0485
sex × age_group	male / 80+	324	-0.452	1.037	0.086	0.0465	0.0450
sex × frailty	Overall	6051	-0.579 (-0.673–0.489)	1.441 (1.325–1.578)	0.062 (0.058–0.066)	0.0472	0.0470
sex × frailty	female / frail	1010	-0.409 (-0.599–0.235)	1.503 (1.255–1.810)	0.083 (0.073–0.093)	0.0547	0.0447
sex × frailty	female / non_frail	275	-3.223	4.169	0.013	0.0722	0.0722
sex × frailty	female / pre_frail	805	-0.540 (-0.835–0.288)	1.065	0.063 (0.051–0.075)	0.0414	0.0409
sex × frailty	male / frail	2674	-0.451 (-0.581–0.329)	1.490 (1.327–1.675)	0.072 (0.066–0.078)	0.0437	0.0448
sex × frailty	male / non_frail	272	-1.346	1.284	0.021	0.0453	0.0453
sex × frailty	male / pre_frail	1015	-1.189	1.145	0.039	0.0654	0.0647
age_group × frailty	Overall	6051	-0.579 (-0.673–0.489)	1.441 (1.325–1.578)	0.062 (0.058–0.066)	0.0472	0.0470
age_group × frailty	0-17 / frail	128	-0.175	1.351	0.111	0.0955	0.0462
age_group × frailty	0-17 / non_frail	28	—	—	0.007	0.0732	0.0732
age_group × frailty	0-17 / pre_frail	67	-1.355	0.199	0.038	0.0914	0.0759
age_group × frailty	18-44 / frail	183	-0.611	1.462	0.084	0.0654	0.0679
age_group × frailty	18-44 / non_frail	182	-2.074	0.855	0.020	0.0637	0.0641
age_group × frailty	18-44 / pre_frail	153	-0.459	0.788	0.067	0.0608	0.0366

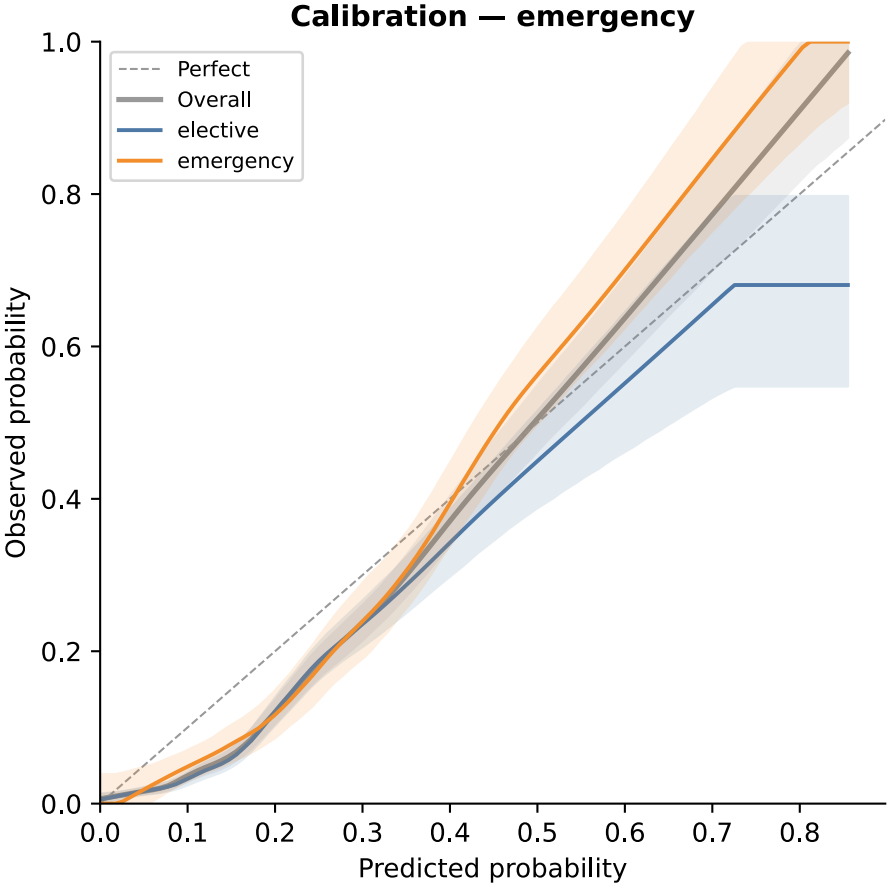
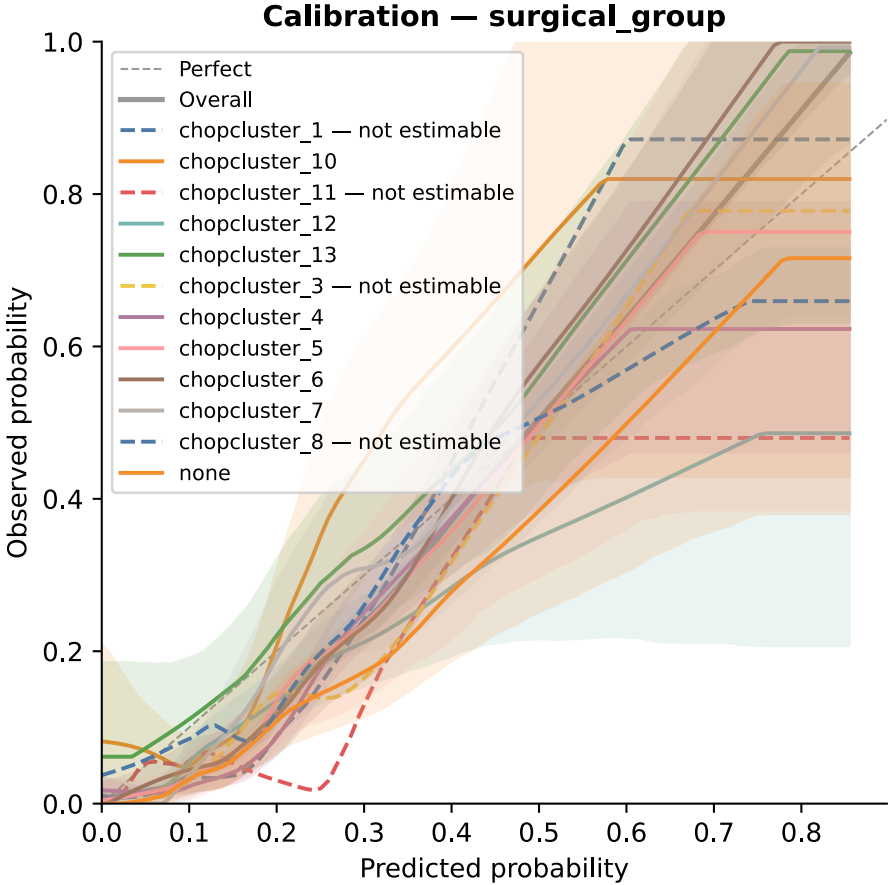
Attribute	Subgroup	n	Intercept (95% CI)	Slope (95% CI)	Brier (95% CI)	ECE	ICI
age_group × frailty	45-64 / frail	1167	-0.588 (-0.800–0.416)	1.495 (1.263–1.786)	0.067 (0.058–0.075)	0.0519	0.0501
age_group × frailty	45-64 / non_frail	210	-1.656	2.705	0.016	0.0510	0.0506
age_group × frailty	45-64 / pre_frail	672	-0.893	1.056	0.048	0.0597	0.0550
age_group × frailty	65-79 / frail	1846	-0.352 (-0.502–0.209)	1.667 (1.449–1.935)	0.071 (0.064–0.079)	0.0559	0.0517
age_group × frailty	65-79 / non_frail	118	-2.196	0.132	0.016	0.0667	0.0588
age_group × frailty	65-79 / pre_frail	734	-0.790	1.311	0.050	0.0533	0.0514
age_group × frailty	80+ / frail	360	-0.445	0.955	0.103	0.0610	0.0520
age_group × frailty	80+ / non_frail	9	—	—	0.020	0.1089	0.1089
age_group × frailty	80+ / pre_frail	194	-1.123	0.817	0.046	0.0726	0.0670

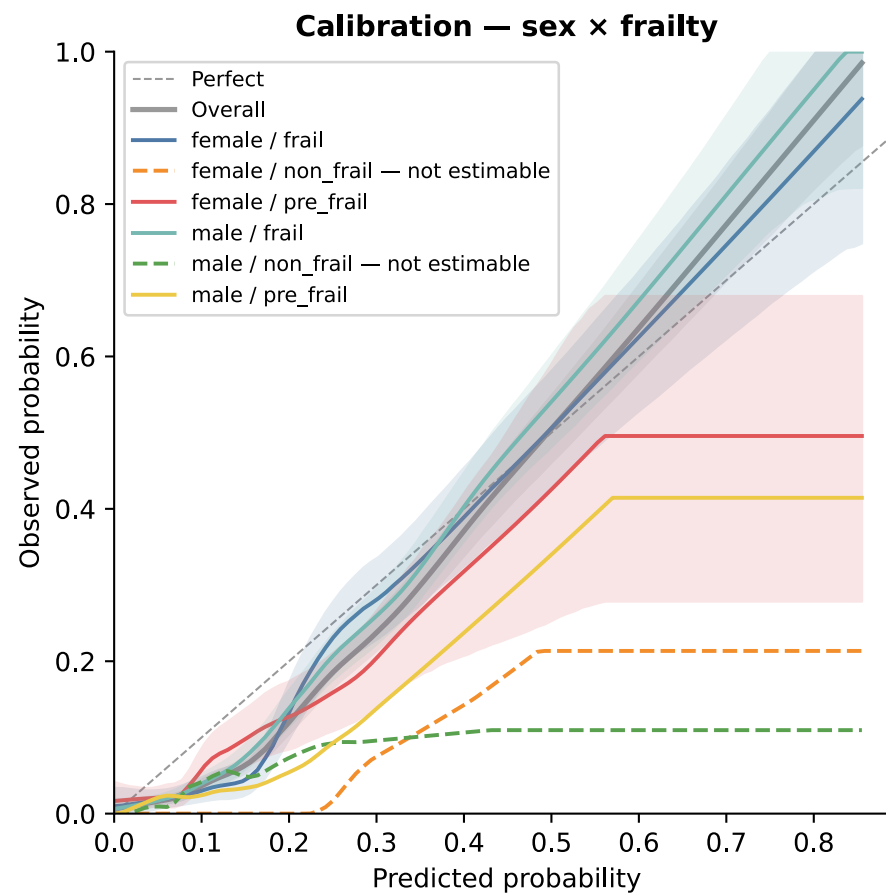
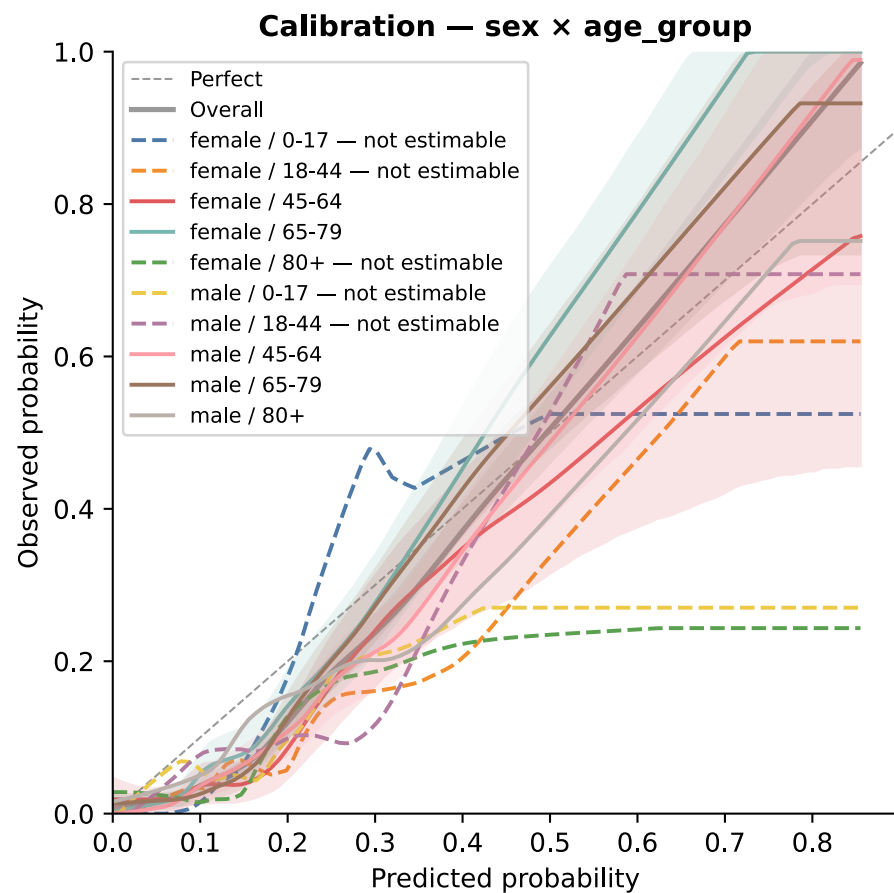
Calibration gap statistics

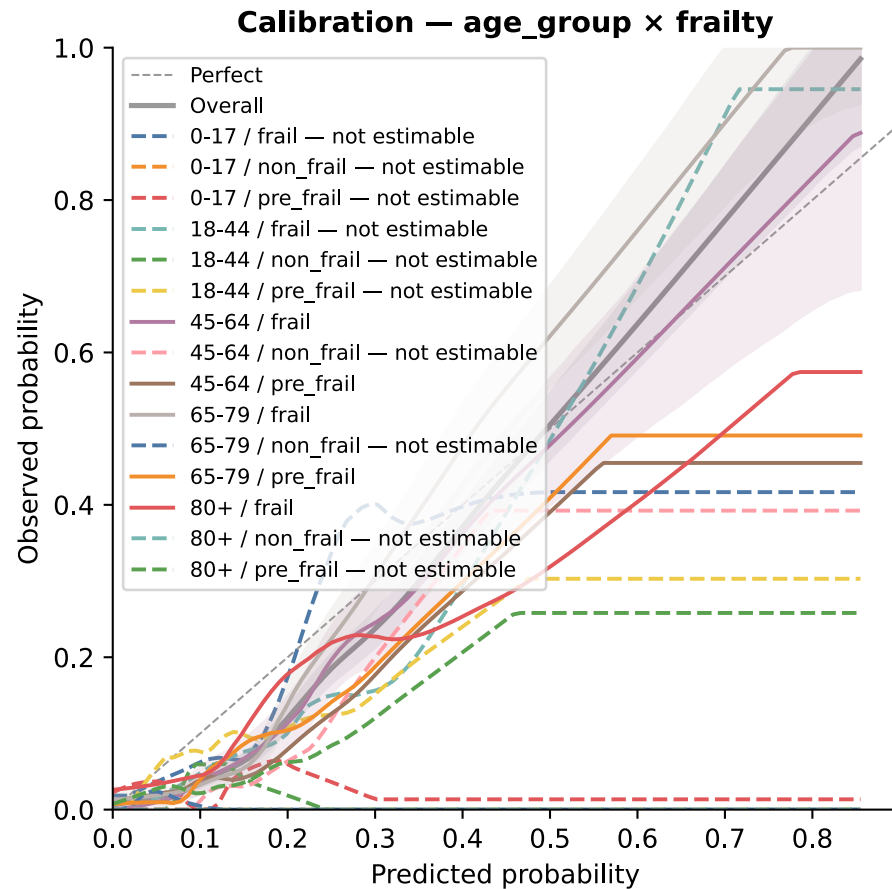
Attribute	Metric	Value (95% CI)
sex	max_intercept_gap	0.0612 (0.0037–0.2505)
sex	max_slope_gap	0.0738 (0.0049–0.3282)
age_group	max_intercept_gap	0.2983
age_group	max_slope_gap	0.6005
frailty	max_intercept_gap	1.5578
frailty	max_slope_gap	0.3782
clinic	max_intercept_gap	0.9065
clinic	max_slope_gap	1.3186
surgical_group	max_intercept_gap	1.0715
surgical_group	max_slope_gap	0.8462
emergency	max_intercept_gap	0.4462 (0.2359–0.6345)
emergency	max_slope_gap	0.2188 (0.0141–0.5215)
sex × age_group	max_intercept_gap	0.6421
sex × age_group	max_slope_gap	1.3029
sex × frailty	max_intercept_gap	2.8143
sex × frailty	max_slope_gap	3.1041
age_group × frailty	max_intercept_gap	2.0208
age_group × frailty	max_slope_gap	2.5730











Hosmer–Lemeshow goodness-of-fit

Read every statistic beside its n and its decile table. The Hosmer–Lemeshow statistic scales with subgroup size: a clinically irrelevant deviation becomes significant in a large subgroup, while a large deviation can go undetected in a small one. Values are therefore **not comparable across subgroups of different size**, and subgroups must not be ranked by p-value. This is why Kramer & Zimmerman (2007) require the statistic to be reported together with the number of patients, the per-decile observed-versus-expected table below, and the adjunct calibration measures in the table above — never on its own.

Attribute	Subgroup	n	Events	H–L statistic	df	p	Groups used	Decile slope	Decile intercept
sex	Overall	6051	501	183.31	10	0.0000	10	0.953	-0.041
sex	female	2090	189	67.06	10	0.0000	10	0.954	-0.038
sex	male	3961	312	119.89	10	0.0000	10	0.941	-0.040
age_group	Overall	6051	501	183.31	10	0.0000	10	0.953	-0.041

Attribute	Subgroup	n	Events	H–L statistic	df	p	Groups used	Decile slope	Decile intercept
age_group	0-17	223	21	8.21	10	0.6080	10	1.177	-0.069
age_group	18-44	518	34	22.24	10	0.0139	10	0.688	-0.017
age_group	45-64	2049	146	77.03	10	0.0000	10	0.887	-0.039
age_group	65-79	2698	247	83.39	10	0.0000	10	1.092	-0.052
age_group	80+	563	53	21.52	10	0.0178	10	0.655	-0.005
frailty	Overall	6051	501	183.31	10	0.0000	10	0.953	-0.041
frailty	frail	3684	399	107.70	10	0.0000	10	1.074	-0.053
frailty	non_frail	547	6	32.12	10	0.0004	10	0.191	-0.002
frailty	pre_frail	1820	96	68.45	10	0.0000	10	0.621	-0.014
clinic	Overall	6051	501	183.31	10	0.0000	10	0.953	-0.041
clinic	ORL	118	14	9.95	10	0.4450	10	0.558	-0.001
clinic	gynecology	274	24	5.05	10	0.8881	10	0.972	-0.017
clinic	neuro	488	25	20.45	10	0.0252	10	0.954	-0.040
clinic	ortho	971	76	28.97	10	0.0013	10	0.736	-0.017
clinic	pediatric	37	3	7.64	10	0.6643	10	0.655	-0.009
clinic	thoracic	403	14	17.59	10	0.0624	10	0.749	-0.029
clinic	urology	200	19	14.04	10	0.1714	10	1.140	-0.042
clinic	vascular	2838	183	124.01	10	0.0000	10	0.959	-0.050
clinic	visceral	722	143	21.18	10	0.0198	10	1.185	-0.064
surgical_group	Overall	6051	501	183.31	10	0.0000	10	0.953	-0.041
surgical_group	chopcluster_1	395	21	20.48	10	0.0250	10	0.910	-0.039
surgical_group	chopcluster_10	186	28	17.78	10	0.0588	10	1.473	-0.054
surgical_group	chopcluster_11	178	9	7.09	10	0.7168	10	0.486	+0.005
surgical_group	chopcluster_12	826	61	24.05	10	0.0075	10	0.756	-0.019
surgical_group	chopcluster_13	154	43	6.23	10	0.7958	10	1.199	-0.021
surgical_group	chopcluster_3	414	22	14.71	10	0.1429	10	0.821	-0.030
surgical_group	chopcluster_4	2245	118	106.33	10	0.0000	10	0.847	-0.041
surgical_group	chopcluster_5	328	30	12.75	10	0.2382	10	0.927	-0.041
surgical_group	chopcluster_6	373	66	22.68	10	0.0120	10	1.277	-0.086
surgical_group	chopcluster_7	143	36	8.13	10	0.6161	10	1.309	-0.094
surgical_group	chopcluster_8	142	23	6.75	10	0.7490	10	1.018	-0.019
surgical_group	none	667	44	34.56	10	0.0001	10	0.734	-0.033
emergency	Overall	6051	501	183.31	10	0.0000	10	0.953	-0.041

Attribute	Subgroup	n	Events	H–L statistic	df	p	Groups used	Decile slope	Decile intercept
emergency	elective	5162	314	168.08	10	0.0000	10	0.863	-0.033
emergency	emergency	889	187	38.27	10	0.0000	10	1.299	-0.115
sex × age_group	Overall	6051	501	183.31	10	0.0000	10	0.953	-0.041
sex × age_group	female / 0-17	109	13	6.09	10	0.8072	10	1.533	-0.116
sex × age_group	female / 18-44	220	13	13.34	10	0.2054	10	0.559	-0.011
sex × age_group	female / 45-64	683	52	27.09	10	0.0025	10	0.846	-0.032
sex × age_group	female / 65-79	839	92	26.09	10	0.0036	10	1.195	-0.058
sex × age_group	female / 80+	239	19	17.46	10	0.0647	10	0.580	-0.009
sex × age_group	male / 0-17	114	8	8.25	10	0.6041	10	0.636	-0.011
sex × age_group	male / 18-44	298	21	8.91	10	0.5404	10	0.674	-0.008
sex × age_group	male / 45-64	1366	94	53.20	10	0.0000	10	0.912	-0.044
sex × age_group	male / 65-79	1859	155	63.70	10	0.0000	10	1.058	-0.051
sex × age_group	male / 80+	324	34	9.05	10	0.5274	10	0.790	-0.013
sex × frailty	Overall	6051	501	183.31	10	0.0000	10	0.953	-0.041
sex × frailty	female / frail	1010	129	36.66	10	0.0001	10	1.083	-0.058
sex × frailty	female / non_frail	275	1	22.33	10	0.0135	10	0.123	-0.006
sex × frailty	female / pre_frail	805	59	19.60	10	0.0333	10	0.737	-0.011
sex × frailty	male / frail	2674	270	73.34	10	0.0000	10	1.068	-0.051
sex × frailty	male / non_frail	272	5	11.06	10	0.3528	10	0.422	-0.009
sex × frailty	male / pre_frail	1015	37	55.10	10	0.0000	10	0.475	-0.012
age_group × frailty	Overall	6051	501	183.31	10	0.0000	10	0.953	-0.041
age_group × frailty	0-17 / frail	128	19	12.18	10	0.2731	10	1.194	-0.048
age_group × frailty	0-17 / non_frail	28	0	—	—	—	10	—	—
age_group × frailty	0-17 / pre_frail	67	2	7.26	10	0.7010	10	0.115	+0.019
age_group × frailty	18-44 / frail	183	21	8.74	10	0.5568	10	0.911	-0.052
age_group × frailty	18-44 / non_frail	182	2	12.67	10	0.2427	10	0.088	+0.005
age_group × frailty	18-44 / pre_frail	153	11	8.17	10	0.6121	10	0.578	+0.010
age_group × frailty	45-64 / frail	1167	110	42.03	10	0.0000	10	0.992	-0.050
age_group × frailty	45-64 / non_frail	210	3	9.95	10	0.4448	10	0.420	-0.013
age_group × frailty	45-64 / pre_frail	672	33	28.53	10	0.0015	10	0.584	-0.012
age_group × frailty	65-79 / frail	1846	204	69.95	10	0.0000	10	1.239	-0.067
age_group × frailty	65-79 / non_frail	118	1	8.53	10	0.5773	10	-0.029	+0.010
age_group × frailty	65-79 / pre_frail	734	42	30.54	10	0.0007	10	0.720	-0.022

Attribute	Subgroup	n	Events	H–L statistic	df	p	Groups used	Decile slope	Decile intercept
<i>age_group</i> × <i>frailty</i>	<i>80+ / frail</i>	360	45	15.15	10	0.1268	10	0.683	+0.007
<i>age_group</i> × <i>frailty</i>	<i>80+ / non_frail</i>	9	0	—	—	—	9 †	—	—
<i>age_group</i> × <i>frailty</i>	<i>80+ / pre_frail</i>	194	8	13.21	10	0.2121	10	0.369	+0.001

† Fewer risk groups than requested were usable, because the subgroup was small or its predicted probabilities were tied. Degrees of freedom follow the reduced count. A well-calibrated model gives a decile slope near 1 and a decile intercept near 0 (Kramer & Zimmerman 2007). 73 tests are reported here; no multiplicity adjustment has been applied. Adjust within an attribute if needed — do not pool the “Overall” rows with the subgroup rows.

Attribute	Metric	Value
sex	max_hl_gap	52.84
age_group	max_hl_gap	75.17
frailty	max_hl_gap	75.58
clinic	max_hl_gap	118.96
surgical_group	max_hl_gap	100.10
emergency	max_hl_gap	129.80
sex × age_group	max_hl_gap	57.61
sex × frailty	max_hl_gap	62.28
age_group × frailty	max_hl_gap	62.69

The gap is the largest pairwise difference in the statistic across subgroups. It inherits the sample-size dependence in full and should be read as a descriptive summary, not as evidence of differential calibration.

Observed versus expected events per decile

Deciles of predicted risk, of approximately equal size, with cut points snapped to tied predicted probabilities so that identical predictions always share a decile.

sex — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	672	4	8.3	0.0060	0.0123	-4.3
2	457	6	12.6	0.0131	0.0275	-6.6
3	627	11	25.0	0.0175	0.0398	-14.0
4	718	11	41.3	0.0153	0.0575	-30.3
5	603	14	46.7	0.0232	0.0775	-32.7
6	562	15	55.7	0.0267	0.0992	-40.7
7	637	40	83.4	0.0628	0.1309	-43.4
8	582	38	103.7	0.0653	0.1782	-65.7
9	591	116	148.9	0.1963	0.2519	-32.9
10	602	246	261.2	0.4086	0.4339	-15.2

sex — female

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	180	3	1.9	0.0167	0.0103	+1.1
2	213	1	5.4	0.0047	0.0254	-4.4
3	212	4	8.5	0.0189	0.0399	-4.5
4	247	3	14.2	0.0121	0.0575	-11.2
5	199	6	15.4	0.0302	0.0772	-9.4
6	219	4	22.2	0.0183	0.1013	-18.2
7	185	16	25.1	0.0865	0.1355	-9.1
8	222	14	41.5	0.0631	0.1869	-27.5
9	204	49	55.3	0.2402	0.2712	-6.3
10	209	89	95.7	0.4258	0.4579	-6.7

sex — male

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	431	1	5.2	0.0023	0.0121	-4.2
2	305	5	8.4	0.0164	0.0275	-3.4
3	415	7	16.5	0.0169	0.0398	-9.5
4	471	8	27.1	0.0170	0.0575	-19.1
5	404	8	31.4	0.0198	0.0776	-23.4
6	375	12	37.2	0.0320	0.0993	-25.2
7	383	22	49.1	0.0574	0.1283	-27.1
8	377	26	64.2	0.0690	0.1702	-38.2
9	413	70	99.6	0.1695	0.2412	-29.6
10	387	153	163.0	0.3953	0.4211	-10.0

age_group — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	672	4	8.3	0.0060	0.0123	-4.3
2	457	6	12.6	0.0131	0.0275	-6.6
3	627	11	25.0	0.0175	0.0398	-14.0
4	718	11	41.3	0.0153	0.0575	-30.3
5	603	14	46.7	0.0232	0.0775	-32.7
6	562	15	55.7	0.0267	0.0992	-40.7
7	637	40	83.4	0.0628	0.1309	-43.4
8	582	38	103.7	0.0653	0.1782	-65.7
9	591	116	148.9	0.1963	0.2519	-32.9
10	602	246	261.2	0.4086	0.4339	-15.2

age_group — 0-17

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	24	0	0.6	0.0000	0.0262	-0.6
2	23	1	1.2	0.0435	0.0513	-0.2
3	22	1	1.5	0.0455	0.0702	-0.5
4	23	1	2.0	0.0435	0.0870	-1.0
5	21	0	2.2	0.0000	0.1040	-2.2
6	25	1	3.2	0.0400	0.1262	-2.2
7	17	1	2.5	0.0588	0.1488	-1.5
8	23	2	4.2	0.0870	0.1817	-2.2
9	23	5	5.2	0.2174	0.2278	-0.2
10	22	9	8.1	0.4091	0.3668	+0.9

age_group — 18-44

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	50	0	0.5	0.0000	0.0092	-0.5
2	45	0	1.2	0.0000	0.0256	-1.2
3	67	0	2.6	0.0000	0.0396	-2.6
4	40	1	2.1	0.0250	0.0532	-1.1
5	57	4	3.8	0.0702	0.0661	+0.2
6	57	1	5.1	0.0175	0.0890	-4.1
7	46	4	5.4	0.0870	0.1172	-1.4
8	54	5	8.5	0.0926	0.1579	-3.5
9	51	3	12.2	0.0588	0.2402	-9.2
10	51	16	21.3	0.3137	0.4176	-5.3

age_group — 45-64

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	206	2	1.9	0.0097	0.0093	+0.1
2	232	2	5.8	0.0086	0.0252	-3.8
3	152	3	5.7	0.0197	0.0373	-2.7
4	247	0	12.8	0.0000	0.0520	-12.8
5	180	1	12.5	0.0056	0.0696	-11.5
6	227	6	20.8	0.0264	0.0915	-14.8
7	203	13	25.1	0.0640	0.1235	-12.1
8	197	12	33.1	0.0609	0.1681	-21.1
9	201	27	48.2	0.1343	0.2400	-21.2
10	204	80	88.7	0.3922	0.4350	-8.7

age_group — 65-79

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	288	2	3.7	0.0069	0.0127	-1.7
2	211	3	5.8	0.0142	0.0276	-2.8
3	280	6	11.2	0.0214	0.0399	-5.2
4	309	5	17.8	0.0162	0.0577	-12.8
5	271	3	21.1	0.0111	0.0780	-18.1
6	267	8	26.3	0.0300	0.0987	-18.3
7	276	17	36.2	0.0616	0.1310	-19.2
8	253	15	44.6	0.0593	0.1762	-29.6
9	277	62	69.9	0.2238	0.2525	-7.9
10	266	126	117.2	0.4737	0.4406	+8.8

age_group — 80+

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	59	1	1.0	0.0169	0.0171	-0.0
2	49	2	1.7	0.0408	0.0342	+0.3
3	71	3	3.8	0.0423	0.0531	-0.8
4	53	0	3.8	0.0000	0.0724	-3.8
5	49	2	4.6	0.0408	0.0938	-2.6
6	58	3	7.2	0.0517	0.1236	-4.2
7	55	4	9.3	0.0727	0.1695	-5.3
8	59	12	13.2	0.2034	0.2243	-1.2
9	56	12	16.3	0.2143	0.2913	-4.3
10	54	14	23.9	0.2593	0.4425	-9.9

frailty — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	672	4	8.3	0.0060	0.0123	-4.3
2	457	6	12.6	0.0131	0.0275	-6.6
3	627	11	25.0	0.0175	0.0398	-14.0
4	718	11	41.3	0.0153	0.0575	-30.3
5	603	14	46.7	0.0232	0.0775	-32.7
6	562	15	55.7	0.0267	0.0992	-40.7
7	637	40	83.4	0.0628	0.1309	-43.4
8	582	38	103.7	0.0653	0.1782	-65.7
9	591	116	148.9	0.1963	0.2519	-32.9
10	602	246	261.2	0.4086	0.4339	-15.2

frailty — frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	368	4	6.2	0.0109	0.0169	-2.2
2	351	6	12.3	0.0171	0.0350	-6.3
3	416	7	21.9	0.0168	0.0527	-14.9
4	383	7	27.6	0.0183	0.0721	-20.6
5	342	7	31.3	0.0205	0.0916	-24.3
6	357	18	41.8	0.0504	0.1171	-23.8
7	347	22	53.0	0.0634	0.1527	-31.0
8	387	44	78.1	0.1137	0.2019	-34.1
9	372	110	107.3	0.2957	0.2885	+2.7
10	361	174	174.0	0.4820	0.4820	-0.0

frailty — non_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	49	0	0.2	0.0000	0.0034	-0.2
2	58	0	0.7	0.0000	0.0125	-0.7
3	56	0	1.3	0.0000	0.0225	-1.3
4	61	0	2.0	0.0000	0.0326	-2.0
5	48	0	2.1	0.0000	0.0428	-2.1
6	46	1	2.4	0.0217	0.0532	-1.4
7	62	0	4.0	0.0000	0.0641	-4.0
8	63	1	5.7	0.0159	0.0897	-4.7
9	52	2	7.1	0.0385	0.1372	-5.1
10	52	2	12.8	0.0385	0.2462	-10.8

frailty — pre_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	216	2	2.1	0.0093	0.0098	-0.1
2	151	2	3.4	0.0132	0.0228	-1.4
3	154	1	5.0	0.0065	0.0322	-4.0
4	194	6	8.6	0.0309	0.0444	-2.6
5	197	3	12.2	0.0152	0.0620	-9.2
6	168	6	13.8	0.0357	0.0823	-7.8
7	184	3	19.6	0.0163	0.1065	-16.6
8	192	16	27.6	0.0833	0.1436	-11.6
9	185	15	38.8	0.0811	0.2097	-23.8
10	179	42	63.8	0.2346	0.3567	-21.8

clinic — gynecology

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	25	0	0.2	0.0000	0.0098	-0.2
2	29	0	0.7	0.0000	0.0228	-0.7
3	33	1	1.2	0.0303	0.0348	-0.2
4	27	2	1.3	0.0741	0.0494	+0.7
5	22	1	1.4	0.0455	0.0620	-0.4
6	35	2	2.9	0.0571	0.0824	-0.9
7	19	2	2.0	0.1053	0.1063	-0.0
8	31	3	4.3	0.0968	0.1376	-1.3
9	26	2	5.5	0.0769	0.2110	-3.5
10	27	11	10.2	0.4074	0.3761	+0.8

clinic — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	672	4	8.3	0.0060	0.0123	-4.3
2	457	6	12.6	0.0131	0.0275	-6.6
3	627	11	25.0	0.0175	0.0398	-14.0
4	718	11	41.3	0.0153	0.0575	-30.3
5	603	14	46.7	0.0232	0.0775	-32.7
6	562	15	55.7	0.0267	0.0992	-40.7
7	637	40	83.4	0.0628	0.1309	-43.4
8	582	38	103.7	0.0653	0.1782	-65.7
9	591	116	148.9	0.1963	0.2519	-32.9
10	602	246	261.2	0.4086	0.4339	-15.2

clinic — neuro

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	55	1	0.5	0.0182	0.0097	+0.5
2	37	0	0.9	0.0000	0.0231	-0.9
3	50	0	1.6	0.0000	0.0324	-1.6
4	50	0	2.2	0.0000	0.0449	-2.2
5	44	0	2.6	0.0000	0.0590	-2.6
6	63	3	4.9	0.0476	0.0770	-1.9
7	42	0	4.1	0.0000	0.0977	-4.1
8	46	4	5.7	0.0870	0.1248	-1.7
9	52	2	8.9	0.0385	0.1713	-6.9
10	49	15	14.8	0.3061	0.3024	+0.2

clinic — ORL

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	12	0	0.2	0.0000	0.0179	-0.2
2	12	0	0.7	0.0000	0.0625	-0.7
3	10	0	1.0	0.0000	0.0975	-1.0
4	15	2	1.9	0.1333	0.1267	+0.1
5	11	1	1.8	0.0909	0.1668	-0.8
6	13	1	2.8	0.0769	0.2162	-1.8
7	12	2	3.1	0.1667	0.2588	-1.1
8	11	3	3.6	0.2727	0.3259	-0.6
9	11	3	4.3	0.2727	0.3909	-1.3
10	11	2	5.5	0.1818	0.5045	-3.5

clinic — ortho

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	107	0	0.9	0.0000	0.0086	-0.9
2	73	1	1.7	0.0137	0.0231	-0.7
3	117	2	4.3	0.0171	0.0371	-2.3
4	88	1	4.9	0.0114	0.0555	-3.9
5	111	3	8.2	0.0270	0.0740	-5.2
6	92	2	9.3	0.0217	0.1014	-7.3
7	92	10	12.8	0.1087	0.1386	-2.8
8	100	13	18.7	0.1300	0.1873	-5.7
9	96	16	25.1	0.1667	0.2617	-9.1
10	95	28	39.6	0.2947	0.4171	-11.6

clinic — pediatric

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	4	0	0.1	0.0000	0.0213	-0.1
2	4	1	0.2	0.2500	0.0488	+0.8
3	4	0	0.3	0.0000	0.0713	-0.3
4	5	0	0.5	0.0000	0.0950	-0.5
5	3	0	0.3	0.0000	0.1167	-0.3
6	4	0	0.5	0.0000	0.1363	-0.5
7	4	0	0.7	0.0000	0.1737	-0.7
8	3	1	0.7	0.3333	0.2167	+0.3
9	3	0	0.8	0.0000	0.2633	-0.8
10	3	1	1.2	0.3333	0.3917	-0.2

clinic — thoracic

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	47	0	0.3	0.0000	0.0070	-0.3
2	43	0	0.8	0.0000	0.0174	-0.8
3	24	0	0.6	0.0000	0.0250	-0.6
4	41	0	1.3	0.0000	0.0323	-1.3
5	50	0	2.1	0.0000	0.0418	-2.1
6	34	0	1.9	0.0000	0.0553	-1.9
7	41	1	3.0	0.0244	0.0737	-2.0
8	46	0	4.9	0.0000	0.1071	-4.9
9	37	3	6.0	0.0811	0.1609	-3.0
10	40	10	13.8	0.2500	0.3455	-3.8

clinic — urology

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	19	1	0.3	0.0526	0.0142	+0.7
2	21	1	0.7	0.0476	0.0348	+0.3
3	22	0	1.1	0.0000	0.0507	-1.1
4	17	2	1.1	0.1176	0.0624	+0.9
5	20	1	1.5	0.0500	0.0768	-0.5
6	22	0	2.2	0.0000	0.1020	-2.2
7	18	0	2.3	0.0000	0.1283	-2.3
8	21	3	3.3	0.1429	0.1576	-0.3
9	20	1	4.2	0.0500	0.2105	-3.2
10	20	10	7.5	0.5000	0.3730	+2.5

clinic — vascular

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	248	2	3.3	0.0081	0.0132	-1.3
2	322	8	9.7	0.0248	0.0302	-1.7
3	298	2	13.4	0.0067	0.0448	-11.4
4	272	3	16.3	0.0110	0.0600	-13.3
5	319	4	24.8	0.0125	0.0778	-20.8
6	247	5	23.9	0.0202	0.0968	-18.9
7	282	11	34.2	0.0390	0.1212	-23.2
8	291	14	46.4	0.0481	0.1595	-32.4
9	278	29	60.7	0.1043	0.2184	-31.7
10	281	105	104.7	0.3737	0.3725	+0.3

clinic — visceral

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	79	1	1.1	0.0127	0.0145	-0.1
2	61	0	2.2	0.0000	0.0366	-2.2
3	79	3	5.0	0.0380	0.0630	-2.0
4	71	2	6.6	0.0282	0.0928	-4.6
5	75	8	10.5	0.1067	0.1395	-2.5
6	72	7	14.8	0.0972	0.2056	-7.8
7	73	19	19.4	0.2603	0.2653	-0.4
8	69	21	23.2	0.3043	0.3366	-2.2
9	70	26	30.8	0.3714	0.4406	-4.8
10	73	56	45.1	0.7671	0.6173	+10.9

surgical_group — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	672	4	8.3	0.0060	0.0123	-4.3
2	457	6	12.6	0.0131	0.0275	-6.6
3	627	11	25.0	0.0175	0.0398	-14.0
4	718	11	41.3	0.0153	0.0575	-30.3
5	603	14	46.7	0.0232	0.0775	-32.7
6	562	15	55.7	0.0267	0.0992	-40.7
7	637	40	83.4	0.0628	0.1309	-43.4
8	582	38	103.7	0.0653	0.1782	-65.7
9	591	116	148.9	0.1963	0.2519	-32.9
10	602	246	261.2	0.4086	0.4339	-15.2

surgical_group — chopcluster_1

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	33	1	0.2	0.0303	0.0065	+0.8
2	42	0	0.8	0.0000	0.0195	-0.8
3	44	0	1.4	0.0000	0.0327	-1.4
4	44	0	2.0	0.0000	0.0451	-2.0
5	32	0	1.9	0.0000	0.0587	-1.9
6	45	2	3.4	0.0444	0.0763	-1.4
7	37	1	3.6	0.0270	0.0986	-2.6
8	44	2	5.9	0.0455	0.1342	-3.9
9	34	1	6.3	0.0294	0.1844	-5.3
10	40	14	14.0	0.3500	0.3505	-0.0

surgical_group — chopcluster_10

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	16	1	0.3	0.0625	0.0175	+0.7
2	22	2	0.9	0.0909	0.0420	+1.1
3	17	0	1.0	0.0000	0.0571	-1.0
4	21	3	1.5	0.1429	0.0731	+1.5
5	20	0	2.0	0.0000	0.0998	-2.0
6	18	0	2.3	0.0000	0.1264	-2.3
7	17	4	2.6	0.2353	0.1521	+1.4
8	19	1	3.5	0.0526	0.1850	-2.5
9	19	6	4.6	0.3158	0.2408	+1.4
10	17	11	7.2	0.6471	0.4221	+3.8

surgical_group — chopcluster_11

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	18	0	0.2	0.0000	0.0086	-0.2
2	15	0	0.3	0.0000	0.0183	-0.3
3	19	0	0.5	0.0000	0.0271	-0.5
4	21	1	0.8	0.0476	0.0398	+0.2
5	19	2	1.1	0.1053	0.0558	+0.9
6	15	0	1.1	0.0000	0.0713	-1.1
7	22	1	1.9	0.0455	0.0841	-0.9
8	15	1	1.6	0.0667	0.1077	-0.6
9	17	1	2.8	0.0588	0.1674	-1.8
10	17	3	5.7	0.1765	0.3356	-2.7

surgical_group — chopcluster_12

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	97	0	0.8	0.0000	0.0087	-0.8
2	62	1	1.4	0.0161	0.0231	-0.4
3	101	2	3.8	0.0198	0.0373	-1.8
4	73	0	4.1	0.0000	0.0555	-4.1
5	87	2	6.2	0.0230	0.0717	-4.2
6	86	2	8.2	0.0233	0.0953	-6.2
7	78	8	10.0	0.1026	0.1285	-2.0
8	82	9	14.6	0.1098	0.1777	-5.6
9	78	14	19.3	0.1795	0.2480	-5.3
10	82	23	32.5	0.2805	0.3963	-9.5

surgical_group — chopcluster_13

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	16	1	0.8	0.0625	0.0522	+0.2
2	17	3	1.5	0.1765	0.0897	+1.5
3	16	1	1.9	0.0625	0.1200	-0.9
4	16	2	2.7	0.1250	0.1675	-0.7
5	14	2	2.9	0.1429	0.2054	-0.9
6	15	6	3.6	0.4000	0.2400	+2.4
7	16	5	4.8	0.3125	0.2978	+0.2
8	15	6	5.5	0.4000	0.3640	+0.5
9	14	6	6.0	0.4286	0.4279	+0.0
10	15	11	8.8	0.7333	0.5840	+2.2

surgical_group — chopcluster_3

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	45	0	0.3	0.0000	0.0071	-0.3
2	44	0	0.8	0.0000	0.0175	-0.8
3	27	0	0.7	0.0000	0.0250	-0.7
4	66	0	2.3	0.0000	0.0355	-2.3
5	28	0	1.3	0.0000	0.0466	-1.3
6	43	0	2.7	0.0000	0.0617	-2.7
7	41	1	3.6	0.0244	0.0879	-2.6
8	38	2	4.7	0.0526	0.1239	-2.7
9	41	6	8.4	0.1463	0.2057	-2.4
10	41	13	16.9	0.3171	0.4132	-3.9

surgical_group — chopcluster_4

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	193	2	2.6	0.0104	0.0133	-0.6
2	276	7	8.3	0.0254	0.0302	-1.3
3	171	2	7.2	0.0117	0.0424	-5.2
4	237	3	13.0	0.0127	0.0548	-10.0
5	266	2	19.1	0.0075	0.0719	-17.1
6	230	4	21.0	0.0174	0.0914	-17.0
7	204	5	23.4	0.0245	0.1147	-18.4
8	220	9	32.2	0.0409	0.1462	-23.2
9	228	15	45.5	0.0658	0.1996	-30.5
10	220	69	75.2	0.3136	0.3417	-6.2

surgical_group — chopcluster_7

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	14	0	0.6	0.0000	0.0436	-0.6
2	16	0	1.3	0.0000	0.0797	-1.3
3	14	0	1.7	0.0000	0.1225	-1.7
4	16	3	2.8	0.1875	0.1725	+0.2
5	13	2	2.8	0.1538	0.2188	-0.8
6	14	5	3.7	0.3571	0.2621	+1.3
7	14	5	4.3	0.3571	0.3082	+0.7
8	14	4	5.2	0.2857	0.3711	-1.2
9	15	6	7.1	0.4000	0.4723	-1.1
10	13	11	8.4	0.8462	0.6477	+2.6

surgical_group — chopcluster_5

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	33	0	0.5	0.0000	0.0138	-0.5
2	35	1	1.1	0.0286	0.0309	-0.1
3	32	0	1.7	0.0000	0.0527	-1.7
4	35	1	2.6	0.0286	0.0729	-1.6
5	33	0	3.1	0.0000	0.0938	-3.1
6	29	1	3.5	0.0345	0.1193	-2.5
7	32	3	4.9	0.0938	0.1522	-1.9
8	34	3	6.6	0.0882	0.1929	-3.6
9	33	9	8.5	0.2727	0.2573	+0.5
10	32	12	14.5	0.3750	0.4537	-2.5

surgical_group — chopcluster_8

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	16	0	0.2	0.0000	0.0116	-0.2
2	12	1	0.4	0.0833	0.0329	+0.6
3	15	2	0.9	0.1333	0.0583	+1.1
4	16	0	1.3	0.0000	0.0800	-1.3
5	14	1	1.5	0.0714	0.1050	-0.5
6	13	2	1.8	0.1538	0.1381	+0.2
7	15	1	3.0	0.0667	0.2027	-2.0
8	13	3	3.6	0.2308	0.2746	-0.6
9	14	4	5.0	0.2857	0.3575	-1.0
10	14	9	7.9	0.6429	0.5675	+1.1

surgical_group — chopcluster_6

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	42	0	0.9	0.0000	0.0210	-0.9
2	33	1	1.3	0.0303	0.0406	-0.3
3	38	0	2.3	0.0000	0.0609	-2.3
4	37	3	3.2	0.0811	0.0858	-0.2
5	38	3	4.9	0.0789	0.1280	-1.9
6	39	0	7.3	0.0000	0.1863	-7.3
7	34	6	8.0	0.1765	0.2365	-2.0
8	36	11	11.1	0.3056	0.3088	-0.1
9	40	12	16.1	0.3000	0.4021	-4.1
10	36	30	22.3	0.8333	0.6183	+7.7

surgical_group — none

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	72	0	0.7	0.0000	0.0092	-0.7
2	73	0	1.9	0.0000	0.0255	-1.9
3	65	0	2.6	0.0000	0.0398	-2.6
4	64	0	3.7	0.0000	0.0580	-3.7
5	62	1	4.9	0.0161	0.0798	-3.9
6	65	2	6.9	0.0308	0.1055	-4.9
7	68	5	9.5	0.0735	0.1402	-4.5
8	65	3	12.1	0.0462	0.1866	-9.1
9	66	13	17.0	0.1970	0.2581	-4.0
10	67	20	29.8	0.2985	0.4451	-9.8

emergency — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	672	4	8.3	0.0060	0.0123	-4.3
2	457	6	12.6	0.0131	0.0275	-6.6
3	627	11	25.0	0.0175	0.0398	-14.0
4	718	11	41.3	0.0153	0.0575	-30.3
5	603	14	46.7	0.0232	0.0775	-32.7
6	562	15	55.7	0.0267	0.0992	-40.7
7	637	40	83.4	0.0628	0.1309	-43.4
8	582	38	103.7	0.0653	0.1782	-65.7
9	591	116	148.9	0.1963	0.2519	-32.9
10	602	246	261.2	0.4086	0.4339	-15.2

emergency — elective

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	493	4	4.7	0.0081	0.0095	-0.7
2	624	5	15.8	0.0080	0.0254	-10.8
3	417	8	15.6	0.0192	0.0374	-7.6
4	534	10	26.7	0.0187	0.0499	-16.7
5	493	8	31.9	0.0162	0.0648	-23.9
6	528	12	43.6	0.0227	0.0825	-31.6
7	530	16	56.6	0.0302	0.1068	-40.6
8	534	30	77.5	0.0562	0.1452	-47.5
9	503	55	106.3	0.1093	0.2113	-51.3
10	506	166	185.8	0.3281	0.3672	-19.8

emergency — emergency

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	85	2	4.5	0.0235	0.0528	-2.5
2	91	1	8.5	0.0110	0.0936	-7.5
3	94	11	11.9	0.1170	0.1262	-0.9
4	92	7	14.5	0.0761	0.1579	-7.5
5	80	5	15.1	0.0625	0.1882	-10.1
6	91	12	19.9	0.1319	0.2187	-7.9
7	89	20	23.6	0.2247	0.2648	-3.6
8	89	26	30.0	0.2921	0.3368	-4.0
9	91	35	40.4	0.3846	0.4438	-5.4
10	87	68	53.9	0.7816	0.6198	+14.1

sex × age_group — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	672	4	8.3	0.0060	0.0123	-4.3
2	457	6	12.6	0.0131	0.0275	-6.6
3	627	11	25.0	0.0175	0.0398	-14.0
4	718	11	41.3	0.0153	0.0575	-30.3
5	603	14	46.7	0.0232	0.0775	-32.7
6	562	15	55.7	0.0267	0.0992	-40.7
7	637	40	83.4	0.0628	0.1309	-43.4
8	582	38	103.7	0.0653	0.1782	-65.7
9	591	116	148.9	0.1963	0.2519	-32.9
10	602	246	261.2	0.4086	0.4339	-15.2

sex × age_group — female / 0-17

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	9	0	0.2	0.0000	0.0256	-0.2
2	14	0	0.8	0.0000	0.0543	-0.8
3	12	0	0.9	0.0000	0.0775	-0.9
4	11	0	1.1	0.0000	0.0964	-1.1
5	9	0	1.0	0.0000	0.1144	-1.0
6	10	1	1.4	0.1000	0.1430	-0.4
7	12	1	2.1	0.0833	0.1767	-1.1
8	12	3	2.6	0.2500	0.2125	+0.4
9	11	3	2.9	0.2727	0.2609	+0.1
10	9	5	3.7	0.5556	0.4150	+1.3

sex × age_group — female / 18-44

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	21	0	0.2	0.0000	0.0107	-0.2
2	20	0	0.6	0.0000	0.0308	-0.6
3	24	0	1.1	0.0000	0.0442	-1.1
4	26	0	1.5	0.0000	0.0581	-1.5
5	20	2	1.5	0.1000	0.0730	+0.5
6	24	0	2.2	0.0000	0.0933	-2.2
7	19	1	2.2	0.0526	0.1150	-1.2
8	22	2	3.4	0.0909	0.1534	-1.4
9	23	3	6.0	0.1304	0.2598	-3.0
10	21	5	9.4	0.2381	0.4464	-4.4

sex × age_group — female / 45-64

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	84	2	0.9	0.0238	0.0101	+1.1
2	50	1	1.1	0.0200	0.0224	-0.1
3	80	2	2.8	0.0250	0.0345	-0.8
4	60	0	3.0	0.0000	0.0494	-3.0
5	73	1	4.9	0.0137	0.0676	-3.9
6	67	2	5.9	0.0299	0.0881	-3.9
7	67	4	8.0	0.0597	0.1199	-4.0
8	64	2	10.9	0.0312	0.1701	-8.9
9	70	10	17.4	0.1429	0.2486	-7.4
10	68	28	31.5	0.4118	0.4637	-3.5

sex × age_group — female / 65-79

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	89	1	1.2	0.0112	0.0135	-0.2
2	94	0	2.8	0.0000	0.0300	-2.8
3	76	1	3.4	0.0132	0.0447	-2.4
4	73	1	4.4	0.0137	0.0596	-3.4
5	82	2	6.3	0.0244	0.0770	-4.3
6	89	2	9.0	0.0225	0.1011	-7.0
7	86	11	12.2	0.1279	0.1415	-1.2
8	81	6	15.5	0.0741	0.1917	-9.5
9	87	23	23.9	0.2644	0.2751	-0.9
10	82	45	38.6	0.5488	0.4712	+6.4

sex × age_group — female / 80+

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	30	0	0.6	0.0000	0.0183	-0.6
2	19	1	0.6	0.0526	0.0332	+0.4
3	26	2	1.3	0.0769	0.0502	+0.7
4	19	0	1.2	0.0000	0.0645	-1.2
5	26	0	2.3	0.0000	0.0896	-2.3
6	25	0	3.1	0.0000	0.1252	-3.1
7	24	1	4.1	0.0417	0.1690	-3.1
8	24	5	5.8	0.2083	0.2419	-0.8
9	23	5	7.3	0.2174	0.3185	-2.3
10	23	5	10.2	0.2174	0.4454	-5.2

sex × age_group — male / 0-17

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	11	0	0.3	0.0000	0.0232	-0.3
2	14	1	0.6	0.0714	0.0461	+0.4
3	9	0	0.6	0.0000	0.0644	-0.6
4	14	2	1.1	0.1429	0.0807	+0.9
5	12	0	1.2	0.0000	0.0996	-1.2
6	9	1	1.0	0.1111	0.1139	-0.0
7	12	0	1.6	0.0000	0.1321	-1.6
8	11	1	1.7	0.0909	0.1555	-0.7
9	11	0	2.2	0.0000	0.2000	-2.2
10	11	3	3.7	0.2727	0.3341	-0.7

sex × age_group — male / 18-44

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	32	0	0.3	0.0000	0.0092	-0.3
2	31	0	0.8	0.0000	0.0255	-0.8
3	28	0	1.0	0.0000	0.0368	-1.0
4	36	1	1.8	0.0278	0.0507	-0.8
5	26	2	1.7	0.0769	0.0671	+0.3
6	28	1	2.4	0.0357	0.0870	-1.4
7	28	3	3.4	0.1071	0.1198	-0.4
8	32	3	5.2	0.0938	0.1625	-2.2
9	28	3	6.5	0.1071	0.2325	-3.5
10	29	8	11.5	0.2759	0.3962	-3.5

sex × age_group — male / 45-64

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	122	0	1.1	0.0000	0.0087	-1.1
2	153	1	3.9	0.0065	0.0252	-2.9
3	144	1	5.7	0.0069	0.0397	-4.7
4	126	0	6.9	0.0000	0.0544	-6.9
5	153	2	10.9	0.0131	0.0713	-8.9
6	126	2	12.0	0.0159	0.0951	-10.0
7	139	9	17.2	0.0647	0.1240	-8.2
8	128	10	21.1	0.0781	0.1647	-11.1
9	142	18	33.4	0.1268	0.2355	-15.4
10	133	51	56.2	0.3835	0.4222	-5.2

sex × age_group — male / 65-79

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	199	1	2.5	0.0050	0.0123	-1.5
2	148	3	4.1	0.0203	0.0276	-1.1
3	197	5	7.9	0.0254	0.0400	-2.9
4	212	4	12.3	0.0189	0.0579	-8.3
5	189	1	14.8	0.0053	0.0785	-13.8
6	154	4	14.8	0.0260	0.0960	-10.8
7	196	7	24.2	0.0357	0.1233	-17.2
8	197	12	33.1	0.0609	0.1680	-21.1
9	184	36	45.0	0.1957	0.2445	-9.0
10	183	82	77.9	0.4481	0.4258	+4.1

sex × frailty — female / frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	106	1	2.1	0.0094	0.0201	-1.1
2	86	1	3.4	0.0116	0.0395	-2.4
3	115	2	6.6	0.0174	0.0572	-4.6
4	97	4	7.5	0.0412	0.0775	-3.5
5	97	2	9.6	0.0206	0.0989	-7.6
6	109	5	14.4	0.0459	0.1324	-9.4
7	92	4	16.2	0.0435	0.1764	-12.2
8	105	18	24.6	0.1714	0.2345	-6.6
9	103	41	34.9	0.3981	0.3385	+6.1
10	100	51	53.2	0.5100	0.5325	-2.2

sex × age_group — male / 80+

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	29	1	0.5	0.0345	0.0159	+0.5
2	38	1	1.4	0.0263	0.0370	-0.4
3	34	1	1.9	0.0294	0.0574	-0.9
4	28	0	2.1	0.0000	0.0741	-2.1
5	38	2	3.6	0.0526	0.0951	-1.6
6	30	3	3.8	0.1000	0.1278	-0.8
7	30	3	5.2	0.1000	0.1733	-2.2
8	32	7	6.9	0.2188	0.2156	+0.1
9	33	5	9.0	0.1515	0.2718	-4.0
10	32	11	13.8	0.3438	0.4316	-2.8

sex × frailty — female / non_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	31	0	0.2	0.0000	0.0061	-0.2
2	20	0	0.3	0.0000	0.0150	-0.3
3	29	0	0.7	0.0000	0.0226	-0.7
4	35	0	1.2	0.0000	0.0349	-1.2
5	25	0	1.2	0.0000	0.0466	-1.2
6	27	0	1.6	0.0000	0.0581	-1.6
7	28	0	2.0	0.0000	0.0707	-2.0
8	28	0	2.7	0.0000	0.0975	-2.7
9	25	0	3.8	0.0000	0.1520	-3.8
10	27	1	7.3	0.0370	0.2687	-6.3

sex × frailty — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	672	4	8.3	0.0060	0.0123	-4.3
2	457	6	12.6	0.0131	0.0275	-6.6
3	627	11	25.0	0.0175	0.0398	-14.0
4	718	11	41.3	0.0153	0.0575	-30.3
5	603	14	46.7	0.0232	0.0775	-32.7
6	562	15	55.7	0.0267	0.0992	-40.7
7	637	40	83.4	0.0628	0.1309	-43.4
8	582	38	103.7	0.0653	0.1782	-65.7
9	591	116	148.9	0.1963	0.2519	-32.9
10	602	246	261.2	0.4086	0.4339	-15.2

sex × frailty — female / pre_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	91	2	1.0	0.0220	0.0107	+1.0
2	67	1	1.5	0.0149	0.0231	-0.5
3	94	2	3.2	0.0213	0.0345	-1.2
4	71	2	3.5	0.0282	0.0494	-1.5
5	89	1	6.0	0.0112	0.0676	-5.0
6	78	2	6.9	0.0256	0.0887	-4.9
7	73	5	8.6	0.0685	0.1175	-3.6
8	83	11	13.0	0.1325	0.1570	-2.0
9	79	11	17.9	0.1392	0.2269	-6.9
10	80	22	29.8	0.2750	0.3729	-7.8

sex × frailty — male / frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	291	3	4.9	0.0103	0.0170	-1.9
2	262	5	9.2	0.0191	0.0351	-4.2
3	222	4	11.1	0.0180	0.0500	-7.1
4	300	3	20.2	0.0100	0.0672	-17.2
5	265	6	23.0	0.0226	0.0869	-17.0
6	278	14	31.2	0.0504	0.1123	-17.2
7	265	17	38.9	0.0642	0.1469	-21.9
8	261	30	50.4	0.1149	0.1930	-20.4
9	259	63	69.6	0.2432	0.2688	-6.6
10	271	125	122.4	0.4613	0.4516	+2.6

age_group × frailty — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	672	4	8.3	0.0060	0.0123	-4.3
2	457	6	12.6	0.0131	0.0275	-6.6
3	627	11	25.0	0.0175	0.0398	-14.0
4	718	11	41.3	0.0153	0.0575	-30.3
5	603	14	46.7	0.0232	0.0775	-32.7
6	562	15	55.7	0.0267	0.0992	-40.7
7	637	40	83.4	0.0628	0.1309	-43.4
8	582	38	103.7	0.0653	0.1782	-65.7
9	591	116	148.9	0.1963	0.2519	-32.9
10	602	246	261.2	0.4086	0.4339	-15.2

sex × frailty — male / non_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	32	0	0.1	0.0000	0.0036	-0.1
2	24	0	0.3	0.0000	0.0119	-0.3
3	27	0	0.6	0.0000	0.0224	-0.6
4	36	0	1.2	0.0000	0.0325	-1.2
5	21	0	0.9	0.0000	0.0424	-0.9
6	28	1	1.5	0.0357	0.0534	-0.5
7	19	0	1.2	0.0000	0.0626	-1.2
8	31	0	2.5	0.0000	0.0811	-2.5
9	26	2	3.2	0.0769	0.1246	-1.2
10	28	2	5.8	0.0714	0.2082	-3.8

age_group × frailty — 0-17 / frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	12	0	0.4	0.0000	0.0358	-0.4
2	13	1	0.9	0.0769	0.0688	+0.1
3	14	1	1.3	0.0714	0.0921	-0.3
4	13	1	1.5	0.0769	0.1119	-0.5
5	14	0	1.9	0.0000	0.1350	-1.9
6	12	2	1.9	0.1667	0.1583	+0.1
7	15	0	2.9	0.0000	0.1943	-2.9
8	11	3	2.5	0.2727	0.2264	+0.5
9	12	7	3.4	0.5833	0.2867	+3.6
10	12	4	5.1	0.3333	0.4212	-1.1

sex × frailty — male / pre_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	88	0	0.6	0.0000	0.0066	-0.6
2	121	1	2.5	0.0083	0.0203	-1.5
3	88	1	2.8	0.0114	0.0323	-1.8
4	116	3	5.1	0.0259	0.0444	-2.1
5	88	2	5.2	0.0227	0.0595	-3.2
6	112	3	8.9	0.0268	0.0798	-5.9
7	100	1	10.4	0.0100	0.1041	-9.4
8	103	4	14.1	0.0388	0.1372	-10.1
9	102	4	20.4	0.0392	0.2001	-16.4
10	97	18	33.2	0.1856	0.3427	-15.2

age_group × frailty — 0-17 / non_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	3	0	0.1	0.0000	0.0167	-0.1
2	3	0	0.1	0.0000	0.0283	-0.1
3	3	0	0.1	0.0000	0.0433	-0.1
4	3	0	0.2	0.0000	0.0550	-0.2
5	4	0	0.3	0.0000	0.0675	-0.3
6	2	0	0.1	0.0000	0.0750	-0.1
7	2	0	0.2	0.0000	0.0850	-0.2
8	4	0	0.4	0.0000	0.1013	-0.4
9	2	0	0.2	0.0000	0.1225	-0.2
10	2	0	0.4	0.0000	0.1900	-0.4

age_group × frailty — 0-17 / pre_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	8	0	0.2	0.0000	0.0219	-0.2
2	7	0	0.3	0.0000	0.0450	-0.3
3	7	1	0.4	0.1429	0.0586	+0.6
4	6	0	0.4	0.0000	0.0733	-0.4
5	8	0	0.7	0.0000	0.0856	-0.7
6	7	0	0.7	0.0000	0.1029	-0.7
7	6	0	0.7	0.0000	0.1242	-0.7
8	6	0	0.9	0.0000	0.1492	-0.9
9	6	1	1.1	0.1667	0.1783	-0.1
10	6	0	1.4	0.0000	0.2417	-1.4

age_group × frailty — 18-44 / pre_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	19	0	0.2	0.0000	0.0095	-0.2
2	14	0	0.4	0.0000	0.0261	-0.4
3	18	0	0.7	0.0000	0.0406	-0.7
4	13	1	0.7	0.0769	0.0565	+0.3
5	16	3	1.2	0.1875	0.0753	+1.8
6	12	0	1.1	0.0000	0.0908	-1.1
7	15	1	1.6	0.0667	0.1097	-0.6
8	16	1	2.2	0.0625	0.1384	-1.2
9	15	2	3.0	0.1333	0.1973	-1.0
10	15	3	5.0	0.2000	0.3343	-2.0

age_group × frailty — 18-44 / frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	21	0	0.6	0.0000	0.0264	-0.6
2	16	0	0.7	0.0000	0.0456	-0.7
3	19	1	1.2	0.0526	0.0639	-0.2
4	22	1	2.0	0.0455	0.0900	-1.0
5	14	0	1.6	0.0000	0.1154	-1.6
6	18	2	2.6	0.1111	0.1450	-0.6
7	20	2	3.8	0.1000	0.1920	-1.8
8	17	2	4.7	0.1176	0.2747	-2.7
9	18	4	6.6	0.2222	0.3678	-2.6
10	18	9	9.1	0.5000	0.5075	-0.1

age_group × frailty — 45-64 / frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	107	1	1.4	0.0093	0.0130	-0.4
2	115	1	3.5	0.0087	0.0303	-2.5
3	135	0	6.4	0.0000	0.0475	-6.4
4	110	1	7.2	0.0091	0.0655	-6.2
5	125	1	10.4	0.0080	0.0833	-9.4
6	101	6	11.1	0.0594	0.1100	-5.1
7	130	8	18.5	0.0615	0.1426	-10.5
8	114	11	22.1	0.0965	0.1936	-11.1
9	113	26	31.9	0.2301	0.2824	-5.9
10	117	55	58.0	0.4701	0.4959	-3.0

age_group × frailty — 18-44 / non_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	22	0	0.1	0.0000	0.0068	-0.1
2	19	0	0.4	0.0000	0.0208	-0.4
3	21	0	0.7	0.0000	0.0333	-0.7
4	10	0	0.4	0.0000	0.0400	-0.4
5	15	0	0.7	0.0000	0.0470	-0.7
6	27	0	1.6	0.0000	0.0574	-1.6
7	15	0	1.0	0.0000	0.0680	-1.0
8	18	1	1.6	0.0556	0.0914	-0.6
9	17	1	2.3	0.0588	0.1374	-1.3
10	18	0	4.7	0.0000	0.2603	-4.7

age_group × frailty — 45-64 / non_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	24	0	0.1	0.0000	0.0035	-0.1
2	12	0	0.1	0.0000	0.0100	-0.1
3	27	0	0.5	0.0000	0.0169	-0.5
4	27	0	0.8	0.0000	0.0278	-0.8
5	12	0	0.4	0.0000	0.0350	-0.4
6	24	0	1.1	0.0000	0.0444	-1.1
7	21	0	1.3	0.0000	0.0605	-1.3
8	22	0	1.8	0.0000	0.0795	-1.8
9	20	1	2.7	0.0500	0.1350	-1.7
10	21	2	5.1	0.0952	0.2421	-3.1

age_group × frailty — 45-64 / pre_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	78	1	0.7	0.0128	0.0090	+0.3
2	63	1	1.4	0.0159	0.0227	-0.4
3	54	1	1.7	0.0185	0.0324	-0.7
4	80	2	3.6	0.0250	0.0450	-1.6
5	66	0	4.1	0.0000	0.0614	-4.1
6	61	3	5.0	0.0492	0.0820	-2.0
7	70	1	7.5	0.0143	0.1068	-6.5
8	67	5	9.6	0.0746	0.1434	-4.6
9	67	4	14.0	0.0597	0.2095	-10.0
10	66	15	22.8	0.2273	0.3453	-7.8

age_group × frailty — 65-79 / pre_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	60	1	0.4	0.0167	0.0073	+0.6
2	99	0	2.0	0.0000	0.0197	-2.0
3	64	0	2.1	0.0000	0.0321	-2.1
4	74	2	3.2	0.0270	0.0434	-1.2
5	63	1	3.8	0.0159	0.0605	-2.8
6	81	0	6.4	0.0000	0.0794	-6.4
7	68	1	7.2	0.0147	0.1063	-6.2
8	81	10	11.9	0.1235	0.1467	-1.9
9	72	7	15.8	0.0972	0.2191	-8.8
10	72	20	27.6	0.2778	0.3838	-7.6

age_group × frailty — 65-79 / frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	193	3	3.3	0.0155	0.0173	-0.3
2	187	3	6.5	0.0160	0.0350	-3.5
3	164	4	8.2	0.0244	0.0499	-4.2
4	191	4	12.7	0.0209	0.0667	-8.7
5	205	2	17.9	0.0098	0.0871	-15.9
6	160	8	17.5	0.0500	0.1092	-9.5
7	200	10	28.9	0.0500	0.1447	-18.9
8	181	13	35.0	0.0718	0.1932	-22.0
9	181	55	49.5	0.3039	0.2735	+5.5
10	184	102	86.0	0.5543	0.4672	+16.0

age_group × frailty — 80+ / frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	35	1	0.8	0.0286	0.0234	+0.2
2	37	2	1.8	0.0541	0.0474	+0.2
3	35	1	2.3	0.0286	0.0646	-1.3
4	35	1	2.9	0.0286	0.0839	-1.9
5	39	1	4.4	0.0256	0.1135	-3.4
6	38	3	6.0	0.0789	0.1586	-3.0
7	33	8	6.6	0.2424	0.2009	+1.4
8	35	10	8.9	0.2857	0.2537	+1.1
9	37	8	11.9	0.2162	0.3212	-3.9
10	36	10	17.2	0.2778	0.4768	-7.2

age_group × frailty — 65-79 / non_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	14	0	0.0	0.0000	0.0029	-0.0
2	10	0	0.1	0.0000	0.0130	-0.1
3	15	0	0.3	0.0000	0.0220	-0.3
4	7	0	0.2	0.0000	0.0300	-0.2
5	12	0	0.5	0.0000	0.0400	-0.5
6	13	1	0.7	0.0769	0.0527	+0.3
7	13	0	0.9	0.0000	0.0688	-0.9
8	12	0	1.2	0.0000	0.0979	-1.2
9	11	0	1.5	0.0000	0.1323	-1.5
10	11	0	2.5	0.0000	0.2259	-2.5

age_group × frailty — 80+ / non_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	1	0	0.0	0.0000	0.0100	-0.0
2	1	0	0.0	0.0000	0.0150	-0.0
3	1	0	0.0	0.0000	0.0200	-0.0
4	1	0	0.1	0.0000	0.0700	-0.1
5	1	0	0.1	0.0000	0.1000	-0.1
6	1	0	0.1	0.0000	0.1100	-0.1
7	1	0	0.2	0.0000	0.1600	-0.2
8	1	0	0.2	0.0000	0.2350	-0.2
9	1	0	0.3	0.0000	0.2600	-0.3

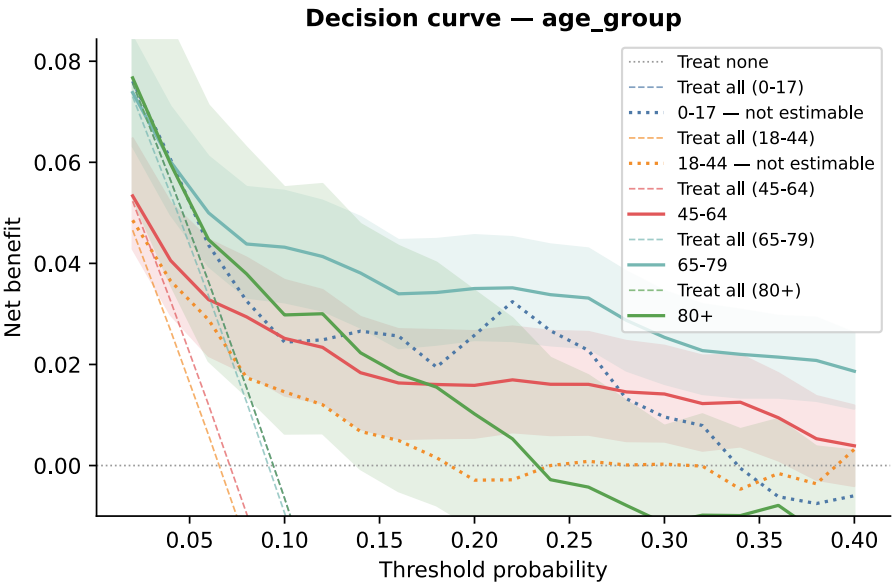
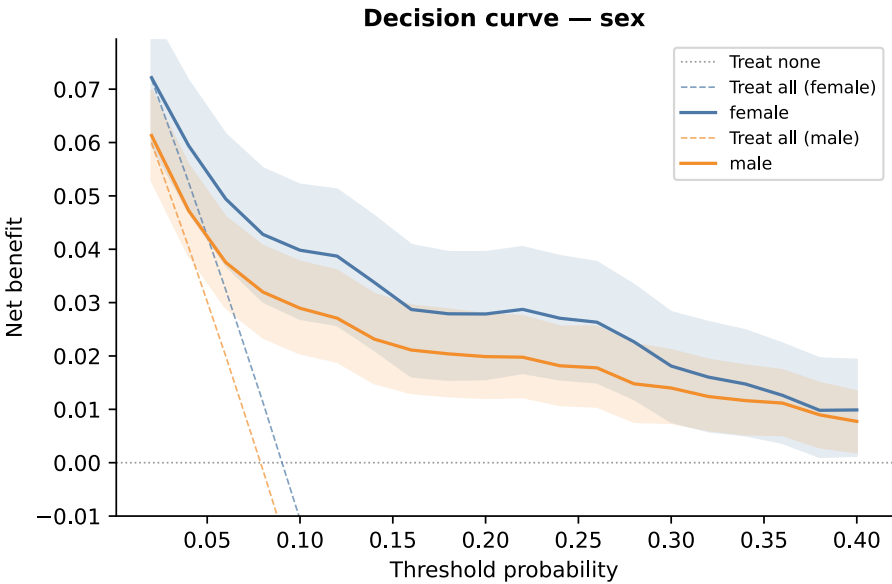
age_group × frailty — 80+ / pre_frail

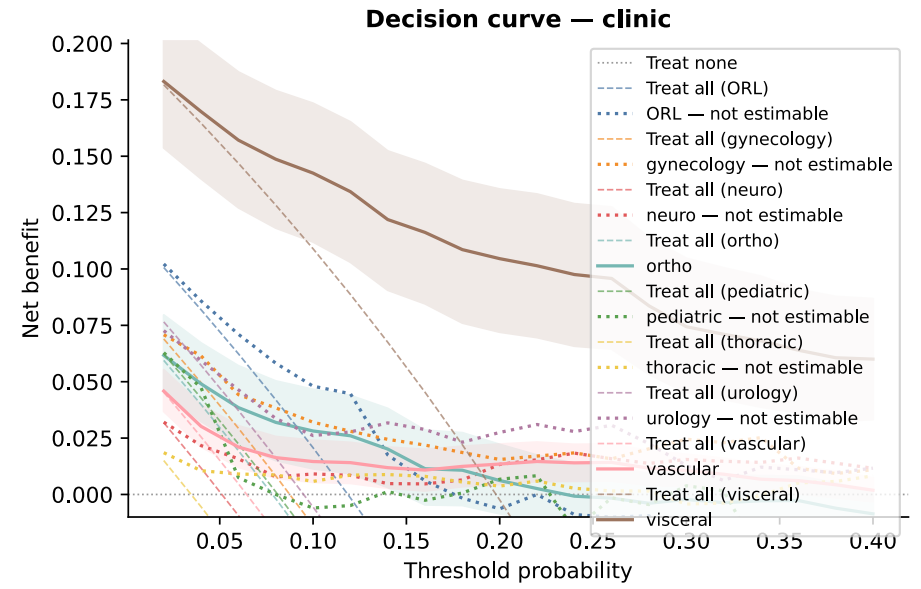
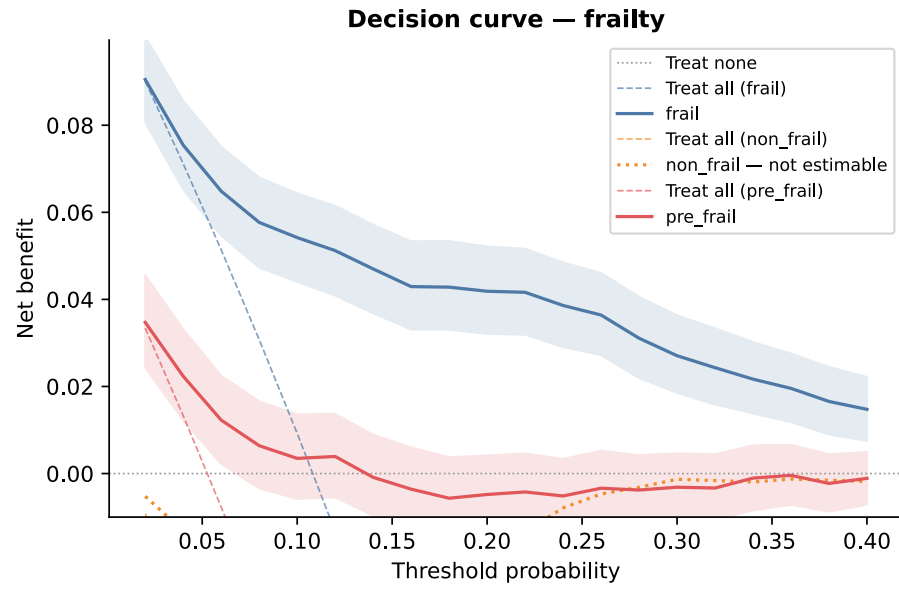
Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	18	0	0.2	0.0000	0.0111	-0.2
2	18	1	0.4	0.0556	0.0236	+0.6
3	25	0	0.8	0.0000	0.0334	-0.8
4	21	1	1.1	0.0476	0.0502	-0.1
5	15	0	1.0	0.0000	0.0647	-1.0
6	21	1	1.8	0.0476	0.0845	-0.8
7	19	0	2.2	0.0000	0.1163	-2.2
8	19	1	2.9	0.0526	0.1550	-1.9
9	19	1	4.3	0.0526	0.2239	-3.3
10	19	3	6.4	0.1579	0.3363	-3.4

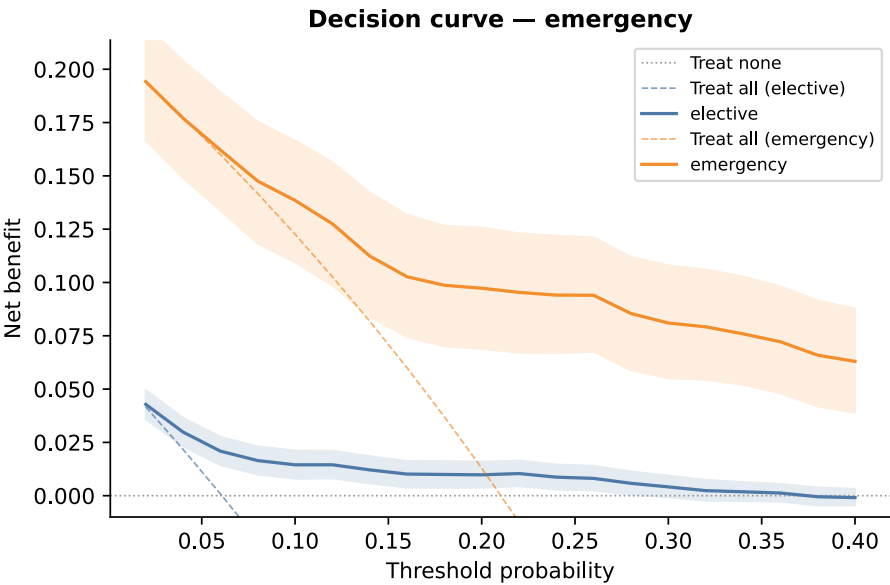
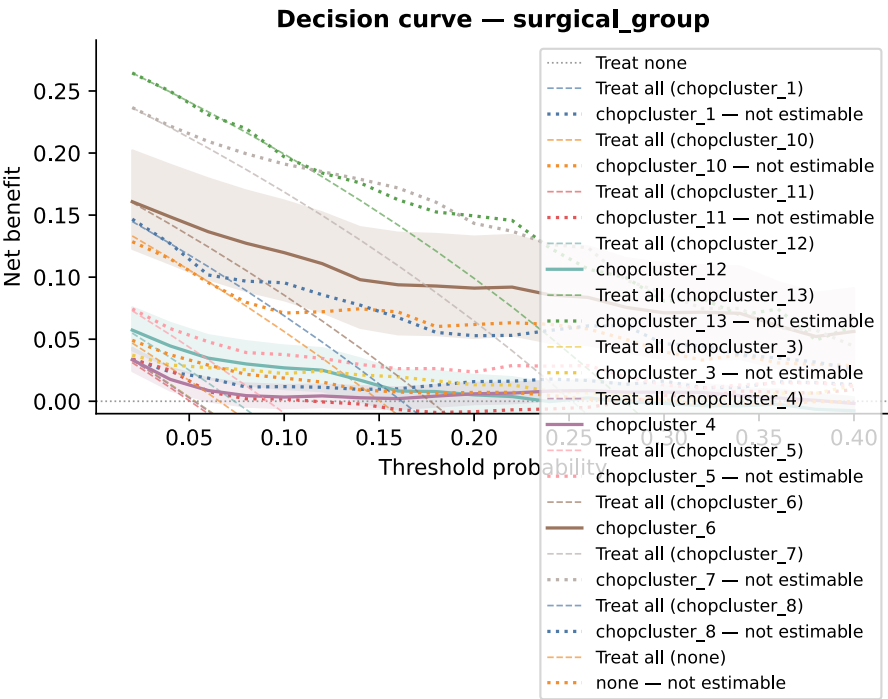
Clinical utility (decision curves)

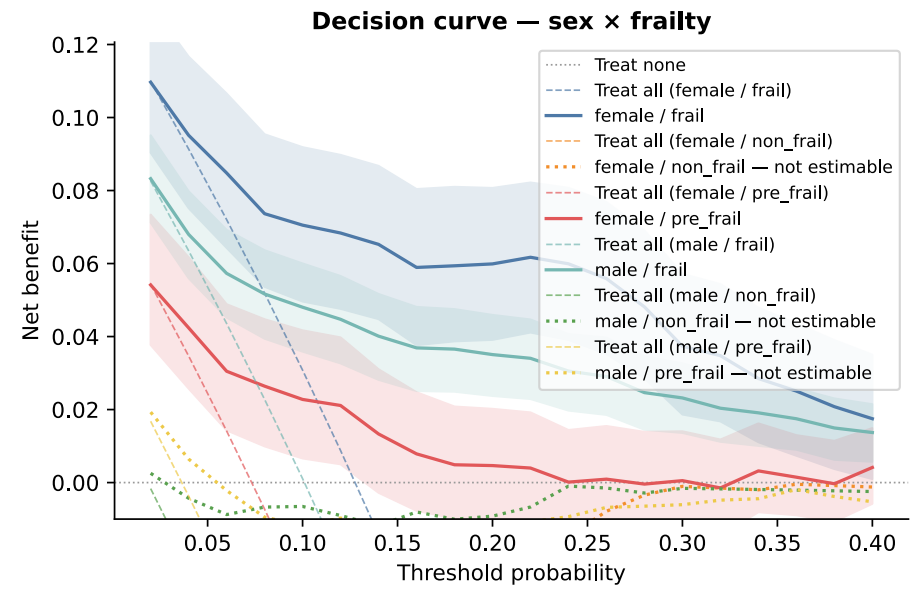
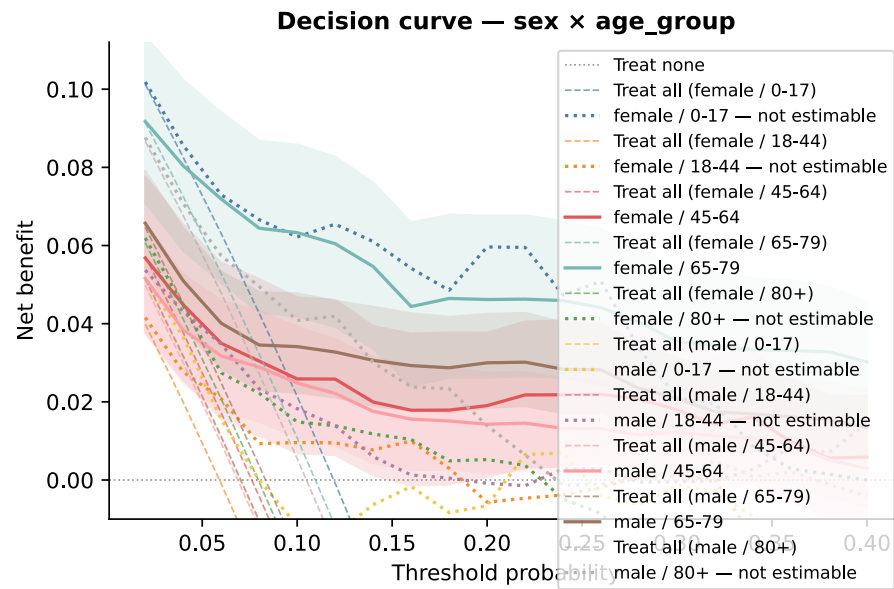
Clinical utility gap statistics

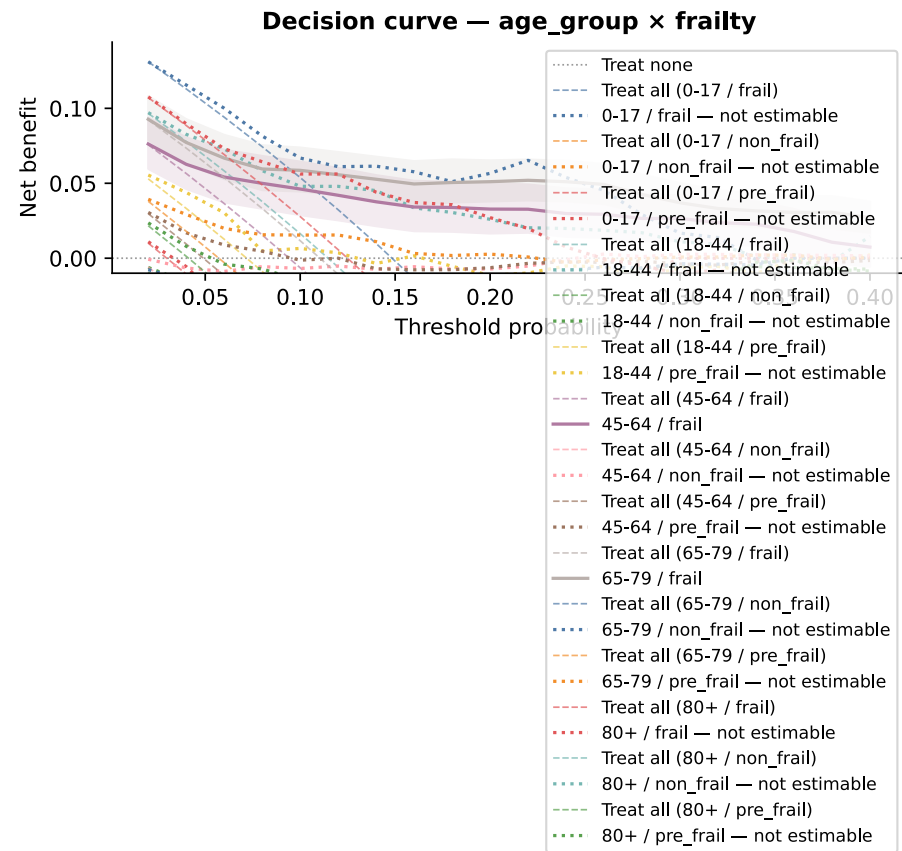
Attribute	Metric	Value (95% CI)
sex	max_net_benefit_gap	0.0122 (0.0052–0.0285)
age_group	max_net_benefit_gap	0.0380
frailty	max_net_benefit_gap	0.0957
clinic	max_net_benefit_gap	0.1646
surgical_group	max_net_benefit_gap	0.2321
emergency	max_net_benefit_gap	0.1514 (0.1247–0.1796)
sex × age_group	max_net_benefit_gap	0.0794
sex × frailty	max_net_benefit_gap	0.1226
age_group × frailty	max_net_benefit_gap	0.1507











Interaction cell diagnostics

Attribute	Subgroup	n	Events	Shrunk	Marginal rate	Cell rate	Shrinkage factor
sex × age_group	female / 0-17	109	13	Yes	0.0904	0.1193	0.6167
sex × age_group	female / 18-44	220	13	Yes	0.0904	0.0591	0.4435
sex × age_group	female / 45-64	683	52	No	0.0904	0.0761	0.2043
sex × age_group	female / 65-79	839	92	No	0.0904	0.1097	0.1729
sex × age_group	female / 80+	239	19	Yes	0.0904	0.0795	0.4232
sex × age_group	male / 0-17	114	8	Yes	0.0788	0.0702	0.6060
sex × age_group	male / 18-44	298	21	Yes	0.0788	0.0705	0.3704
sex × age_group	male / 45-64	1366	94	No	0.0788	0.0688	0.1138
sex × age_group	male / 65-79	1859	155	No	0.0788	0.0834	0.0862
sex × age_group	male / 80+	324	34	Yes	0.0788	0.1049	0.3512
sex × frailty	female / frail	1010	129	No	0.0904	0.1277	0.1713
sex × frailty	female / non_frail	275	1	Yes	0.0904	0.0036	0.4315
sex × frailty	female / pre_frail	805	59	No	0.0904	0.0733	0.2059
sex × frailty	male / frail	2674	270	No	0.0788	0.1010	0.0724
sex × frailty	male / non_frail	272	5	Yes	0.0788	0.0184	0.4342
sex × frailty	male / pre_frail	1015	37	Yes	0.0788	0.0365	0.1706
age_group × frailty	0-17 / frail	128	19	Yes	0.0942	0.1484	0.5107
age_group × frailty	0-17 / non_frail	28	0	Yes	0.0942	0.0000	0.8267
age_group × frailty	0-17 / pre_frail	67	2	Yes	0.0942	0.0299	0.6660
age_group × frailty	18-44 / frail	183	21	Yes	0.0656	0.1148	0.4220
age_group × frailty	18-44 / non_frail	182	2	Yes	0.0656	0.0110	0.4233
age_group × frailty	18-44 / pre_frail	153	11	Yes	0.0656	0.0719	0.4662
age_group × frailty	45-64 / frail	1167	110	No	0.0713	0.0943	0.1027
age_group × frailty	45-64 / non_frail	210	3	Yes	0.0713	0.0143	0.3888
age_group × frailty	45-64 / pre_frail	672	33	Yes	0.0713	0.0491	0.1658
age_group × frailty	65-79 / frail	1846	204	No	0.0915	0.1105	0.0675
age_group × frailty	65-79 / non_frail	118	1	Yes	0.0915	0.0085	0.5310
age_group × frailty	65-79 / pre_frail	734	42	Yes	0.0915	0.0572	0.1540
age_group × frailty	80+ / frail	360	45	Yes	0.0941	0.1250	0.2707
age_group × frailty	80+ / non_frail	9	0	Yes	0.0941	0.0000	0.9369
age_group × frailty	80+ / pre_frail	194	8	Yes	0.0941	0.0412	0.4078

Appendix: TRIPOD+AI item coverage

Fairness-relevant items from the TRIPOD+AI checklist (Collins et al., 2024, *BMJ*). Items addressed by this audit are linked to the relevant report section. Items marked "User responsibility" require information that the audit tool cannot produce automatically.

Item	Description	Status	Detail
3c	Describe any known health inequalities between sociodemographic groups	User responsibility	User must provide context in manuscript introduction
5a	Describe data sources, rationale, and representativeness	User responsibility	User must describe data representativeness in manuscript methods
7	Describe data pre-processing, including whether similar across sociodemographic groups	User responsibility	User must verify preprocessing does not vary by group
8a	Outcome assessment consistency across sociodemographic groups	User responsibility	User must verify outcome labels are measured consistently
8b	Assessor demographics for outcome assessment	User responsibility	User must report assessor demographics
9c	Assessor demographics for predictor measurement	User responsibility	User must report assessor demographics
12e	All measures used to evaluate model performance	Addressed	Discrimination, Calibration, and Clinical utility sections
12f	Describe any model updating (eg recalibration) for sociodemographic groups or settings	Not included	No mitigation requested
14	Fairness approaches and rationale	Addressed	Audit summary and gap statistics throughout
20b	Report demographic characteristics and group differences	Addressed	Discrimination section (subgroup sizes, prevalences)
20c	Compare predictor distributions including demographics	User responsibility	User must report predictor distributions per group in manuscript
23a	Report performance estimates with CIs, including for key sociodemographic subgroups	Addressed	Discrimination, Calibration, and Clinical utility sections
25	Overall interpretation including fairness issues	User responsibility	User must interpret results in manuscript discussion
26	Limitations including non-representative sample, bias effects	User responsibility	User must discuss limitations in manuscript

Source: Collins GS, Moons KGM, Dhiman P, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ* 2024;385:e078378. doi: [10.1136/bmj-2023-078378](https://doi.org/10.1136/bmj-2023-078378). This checklist covers fairness-relevant items only; see the full statement for the complete checklist.

Appendix: How to read this report

Discrimination

Discrimination measures how well the model separates positive from negative cases.

AUROC (Area Under the ROC Curve)

Probability that the model ranks a random positive case higher than a random negative case. Range [0, 1]; 0.5 = no discrimination; 1.0 = perfect separation.

AUPRC (Area Under the Precision-Recall Curve)

Summarises precision across all recall levels. More informative than AUROC for low-prevalence outcomes. Higher is better; baseline equals the prevalence.

Sensitivity (True Positive Rate)

Proportion of true positive cases correctly identified at the chosen threshold. = $TP / (TP + FN)$.

Specificity (True Negative Rate)

Proportion of true negative cases correctly identified at the chosen threshold. = $TN / (TN + FP)$.

PPV (Positive Predictive Value)

Proportion of positive predictions that are truly positive. = $TP / (TP + FP)$.

NPV (Negative Predictive Value)

Proportion of negative predictions that are truly negative. = $TN / (TN + FN)$.

Calibration

Calibration measures whether predicted probabilities match observed event rates.

Calibration intercept

Intercept from logistic recalibration with slope fixed to 1 (Steyerberg 2009). Perfect calibration = 0. Positive values indicate the model underestimates risk on average; negative values indicate overestimation.

Calibration slope

Slope from logistic recalibration with free intercept and slope. Perfect calibration = 1. Values < 1 indicate overconfident predictions (predicted probabilities too extreme); values > 1 indicate underconfident predictions.

Brier score

Mean squared difference between predicted probabilities and binary outcomes. Range [0, 1]; lower is better.

ECE (Expected Calibration Error)

Weighted average of absolute calibration error across equal-frequency bins. Lower is better.

ICI (Integrated Calibration Index)

Mean absolute difference between LOESS-smoothed observed probabilities and predicted probabilities (Austin & Steyerberg 2019). Lower is better; more robust than ECE as it avoids binning artefacts.

Hosmer–Lemeshow statistic

Chi-square-distributed goodness-of-fit statistic comparing observed with expected events across deciles of predicted risk. A large value indicates lack of fit. **It scales with sample size**, so it is not comparable between subgroups of different size and must be read alongside n , the per-decile table, and the calibration measures above (Kramer & Zimmerman 2007).

Hosmer–Lemeshow degrees of freedom

Equal to the number of risk groups used. The classical convention subtracts 2, which is the penalty for estimating an intercept and slope on the data under test; this package evaluates predictions supplied by a model fitted elsewhere, so no degrees of freedom are spent and none are subtracted.

Decile slope and intercept

Least-squares regression of the observed on the expected event rate across risk groups, recommended by Kramer & Zimmerman (2007) as an adjunct to the statistic. Perfect calibration gives a slope of 1 and an intercept of 0.

Clinical utility (decision curves)

Decision curve analysis quantifies the clinical value of using the model at a given decision threshold.

Net benefit

$TP/n - FP/n \times p_t / (1 - p_t)$, where p_t is the decision threshold. Higher is better. A model is clinically useful when its net benefit exceeds both 0 (treat none) and the treat-all reference.

Standardised net benefit

Net benefit divided by prevalence (Naderalvojud 2025). Normalised to a [0, 1] scale for comparison across cohorts with different prevalences.

Treat-all reference

Net benefit if all patients receive treatment. = prevalence – (1 – prevalence) $\times p_t / (1 - p_t)$. Becomes strongly negative at high thresholds.

Treat-none reference

Net benefit = 0. The baseline of no intervention.

Gap statistics

Gap statistics quantify the largest disparity between subgroups for each metric. All gaps are non-negative; 0 = perfect parity.

Equal opportunity difference

max – min of sensitivity across subgroups. Measures fairness in true positive identification.

Predictive equality

max – min of false positive rate (1 – specificity) across subgroups. Measures fairness in false alarm rates.

Equalized odds gap

Maximum of equal opportunity difference and predictive equality.

Statistical parity

max – min of positive prediction rate across subgroups. Measures whether the model predicts positive at equal rates.

Max intercept / slope gap

Maximum pairwise difference in calibration intercept or slope across subgroups, with bootstrap confidence intervals.

Max net benefit gap

Maximum across all thresholds of the maximum pairwise net-benefit difference between subgroups, with bootstrap confidence intervals.

Confidence intervals and shrinkage**Bootstrap confidence intervals**

95% percentile intervals from subgroup-stratified patient-level resamples. Patients are resampled with replacement within each subgroup, preserving the subgroup's size while letting its event count vary.

Empirical Bayes shrinkage

Interaction cells with fewer events than the configured threshold (shown in *italics*) use Beta-Binomial shrinkage toward the marginal subgroup rate. This reduces instability in small cells. Confidence intervals are suppressed for shrunk estimates.

Generated by **isitfair**. Three-axis evaluation: discrimination, calibration, and clinical utility reported jointly per TRIPOD+AI recommendations.

Bootstrap confidence intervals (95%) from 2000 subgroup-stratified patient-level resamples. Interaction cells with fewer than the configured event threshold use empirical Bayes shrinkage (italicised rows).